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Project 25

Digital Land Mobile Radio- Key Fill Device (KFD) Interface Protocol

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FOREWORD

(This foreword is not part of this document)

This document has been created in response to a request by the APCO/NASTD/FED Project 25 Steering Committee as provided for in a Memorandum of Understanding (MOU) dated April 1992 and amended December 1993. The MOU states APCO/NASTD/FED shall devise a Common System Standard for digital public safety communications (the Standard) commonly referred to as Project 25 or P25, and that TIA, as agreed to by the membership of the appropriate TIA Engineering Committee, will provide technical assistance in the development of documentation for the Standard. If the abbreviation "P25" or the wording "Project 25" appears on the cover sheet of this document when published that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted this document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein should not be interpreted as limiting the applicability of the information contained in any document to "P25" or "Project 25" implementations exclusively.

This document has been developed by TR8.3 on Encryption with inputs from the APCO Project 25 Interface Committee (APIC), the APIC Encryption Task Group, and TIA Industry members.

This document is being published because it is felt that there is an urgent need for technical information on the emerging digital techniques for Land Mobile Radio Service.

This document describes procedures and protocols for manual loading of key variables and initial provisioning of key management parameters from a key fill device connected directly to a mobile radio or fixed end equipment. For devices that are not capable of Over The Air Rekeying (OTAR), the Key Fill Device (KFD) is used to provide basic manual rekeying features (loading of keys, erasing keys and viewing key information) and to allow for basic key management by the operator. For OTAR, the KFD is used to provision OTAR parameters (Individual RSI, Key Management Facility RSI, Message Number Period) and to provide the initial Key Encryption Key.

It should be noted that some parts of the Key Fill Device Interface Protocol may not have been operationally tested. All reasonable efforts have been made to ensure accuracy, but it should be understood that conformance tests remain to be developed and proven.

This document includes two normative annexes.

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1 INTRODUCTION

1.1 Purpose

This document describes the protocols, procedures, and features for a manual rekeying function. Manual rekeying involves the exchange of Key Management Messages between a KFD and an MR for the purposes of provisioning an MR with encryption keys, authentication keys or other key management related information, erasing keys in an MR, or viewing key management related information in an MR.

Manual rekeying can be used as the sole means for provisioning mobile radios with traffic encryption keys or it can be used in conjunction with OTAR rekeying procedures. When used with OTAR rekeying procedures, manual rekeying can be used to provision an MR with key management parameters required for OTAR operation in addition to provisioning encryption keys. Manual rekeying can also be used for provisioning mobile radios with authentication keys.

1.2 Scope

This document addresses the manual rekeying interface between a KFD and an MR only. Although some system level overview is provided, system level definition is not fully covered.

1.3 Revision History

January 25, 2011:	TR8.3 11-003, Draft 1
October 18, 2011:	TR8.3 11-022 PN-3-0130-RV1
January 14, 2012:	TR8.3 12-004, PN-3-0130-RV1
March 16, 2012:	TR8.3 12-007, PN-3-0130-RV1
April 20, 2012:	TR8.3 12-007-R1, PN-3-0130-RV1
June 5, 2012:	TR8.3 12-007-R2, PN-3-0130-RV1
June 25, 2012:	TR8.3 12-007-R3, PN-3-0130-RV1
August 29, 2012:	TR8.3 12-007-R4, PN-3-0130-RV1
January 10, 2013:	TR8.3 13-002 PN-3-0130-RV1
April 10, 2013:	TR8.3 13-002-R1 PN-3-0130-RV1
September 04, 2013:	TR8.3 13-002-R2 PN-3-0130-RV1
December 02, 2013:	TR8.3 13-002-R3 PN-3-0130-RV1
May 18, 2014:	TR8.3 13-002-R4 PN-3-0130-RV1
May 30, 2014:	TR8.3 13-002-R5 PN-3-0130-RV1
July 30, 2014	TR8.3 13-002-R6 PN-3-0130-RV1

1.4 References

The following documents contain provisions which, through reference in this text, constitute provisions of this document. At the time of publication, the

editions indicated were valid. All documents are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. ANSI and TIA maintain registers of their currently published documents.

The appearance of “Project 25” in references below indicates the TIA document has been adopted by the APCO/NASTD/FED Project 25 Steering Committee as part of its Project 25 standard, i.e. “the Standard”.

Normative References

- [1] ANSI/TIA-PN-102.AACA-A to be published as TIA/EIA-102.AACA-A, *Over-The-Air-Rekeying (OTAR) Protocol Messages and Procedures*
- [2] TIA-PN-102.BAEG, *Mobile Data Peripheral Interface*, currently under development

Informative References

- [3] *Universal Serial Bus Specification*, Revision 2.0, April 27, 2000.
- [4] RFC 1661 (STD0051), *The Point-to-Point Protocol (PPP)*, July 1994.
- [5] TIA-102.AACE-A, *Project 25 Digital Land Mobile Radio – Link Layer Authentication*, April 2011
- [6] FIPS Publication 197, *Advanced Encryption Standard*, November 26, 2001
- [7] TIA 102.BAJD, *Project 25 TCP/UDP Port Number Assignments*, October 2010
- [8] RFC 1662 (STD0051), *PPP in HDLC-like Framing*, July 1994.
- [9] *USB Class Definitions for Communication Devices*, Version 1.1, January 19, 1999.
- [10] RFC 768, *User Datagram Protocol*, August 1980.
- [11] RFC-791, *Internet Protocol*, IETF (<http://www.rfc-editor.org/>), September 1981

1.5 Definitions, Acronyms, and Abbreviations

\$	Designates a number as a hexadecimal integer
3WI	Three Wire Half Duplex Interface
AES	Advanced Encryption Standard
AF	Authentication Facility
BJT	Bipolar Junction Transistor

CRC	Cyclic Redundancy Code
DES	Data Encryption Standard
K	Authentication Key
kbps	kilo-bits per second
K/F	Key Fill line
KMF	Key Management Facility
KMM	Key Management Message
KFD	Key Fill Device
MNP	Message Number Period
MR	Mobile Radio
OTAR	Over The Air Rekeying
RSI	Radio Set Identifier
SCO	Session Control Opcode
SDT	Source Device Type
SLN	Storage Location Number
SUID	Subscriber Unit Identity
TDES	Triple Data Encryption Standard
TEK	Traffic Encryption Key
UKEK	Unique Key Encryption Key
V_{ILmax}	Voltage Input Low Maximum
V_{IHmin}	Voltage Input High Minimum
V_{OLmax}	Voltage Output Low Maximum

2 MANUAL REKEYING OVERVIEW

For purposes of this section, radio systems are classified as either KFD based or KMF based. KFD based systems are those that rely exclusively on the Key Fill Device for provisioning MRs with encryption keys or authentication keys. KMF based systems utilize a KMF and OTAR procedures in conjunction with a KFD.

2.1 KFD Based Key Management

KFD based key management can be used for systems that utilize encryption for voice or data security or authentication but do not employ a KMF or use OTAR procedures for the distribution of encryption keys. In such systems, a KFD provides the only means for managing encryption keys in the system.

All traffic encryption keys (TEKs) used in a KFD based system are entered and stored in one or more KFDs. The KFDs are then used to load the appropriate TEKs into each of the MRs in the system. As long as two MRs wishing to communicate securely have been loaded with a common TEK, secure communications can take place.

Keeping track of which MRs should be loaded with which TEKs is a manual process and is the responsibility of the KFD operator.

There are four basic manual rekeying features when using a KFD based system. See section 2.3 for a description of these manual rekeying KFD features.

- Keyload
- Key Erase
- Erase All Keys
- View Key Info

These four manual rekeying KFD features provide the basis for managing a simple secure radio system.

All authentication keys (K) used in a KFD based system are entered in or created by a KFD. The KFD is then used to send K to a Subscriber Unit Identity (SUID) in a MR. As long as the K-SUID pair is communicated back to an Authentication Facility (AF), the MR can be authenticated.

There are two basic manual rekeying features when using a KFD based system for authentication. See section 2.4 for a description of these manual rekeying KFD features for authentication.

- Load Authentication Key
- View Active SUID

2.2 KMF Based Key Management

KMF based systems utilize a Key Management Facility for centralized key management. The KMF distributes keys to MRs in the system primarily through the use of OTAR procedures but manual rekeying procedures can also be utilized. Furthermore, manual rekeying procedures are also required for initial provisioning of UKEKs to the MRs as well as provisioning of OTAR related key management parameters to the MRs.

In a KMF based system, each MR in the system is provisioned with four items before OTAR procedures can be used to rekey that MR. These four items are:

- Unique Key Encryption Key (UKEK) per supported algorithm
- Individual Radio Set Identifier (Ind RSI)
- Key Management Facility Radio Set Identifier (KMF RSI)
- Message Number Period (MNP)

Also, ongoing OTAR operations may require a means to activate a keyset in an MR when that MR is not within OTAR coverage.

To support KMF based systems, the following manual rekeying features should be supported by a KFD. See section 2.3 for a description of these manual rekeying KFD features.

- Keyload
- View Individual RSI
- Load Individual RSI
- View KMF RSI
- Load KMF RSI
- View MNP
- Load MNP
- View Keyset Info
- Activate Keyset

2.3 Manual Rekeying Features

2.3.1 Keyload

The Keyload feature of the KFD is used to transfer one or more encryption keys from the KFD to a connected MR for storage in a keyset within the MR. In a KFD based system, this feature is used to transfer one or more TEKs from a KFD to an MR. In KMF based systems, this feature is primarily used to load one or more UKEKs into the MR. However, loading of TEKs may also be needed when the MR is out of OTAR RF coverage area. See section 3.8.1 for a detailed description of the procedures used for implementing this feature.

2.3.2 Key Erase

The Key Erase feature of the KFD is used to erase one or more specific encryption keys within a connected MR. See section 3.8.5 for a detailed description of the procedures used for implementing this feature.

2.3.3 Erase All Keys

The Erase All Keys feature of the KFD is used to erase all KEKs and TEKs stored in a connected MR, those stored in active keysets as well as those stored in inactive keysets. It should be noted that the Erase All Keys feature does not affect the authentication keys, K. See section 3.8.6 for a detailed description of the procedures used for implementing this feature.

2.3.4 View Key Info

The View Key Info feature of the KFD is used to view information about keys stored in a connected MR. The key information to be viewed includes Keyset Ids, Storage Location Numbers, Key IDs, and Algorithm IDs of the keys stored in the connected MR. See section 3.8.7 for a detailed description of the procedures used for implementing this feature.

2.3.5 View Individual RSI

The View Individual RSI feature of the KFD is used to retrieve the RSI of a connected MR. The Individual RSI programmed in the MR is used to determine if OTAR KMMs received via the RF channel are intended for the MR. See section 3.8.8 for a detailed description of the procedures used for implementing this feature.

2.3.6 Load Individual RSI

The Load Individual RSI feature of the KFD is used to set the RSI of a connected MR. See section 3.8.2 for a detailed description of the procedures used for implementing this feature.

2.3.7 View KMF RSI

The View KMF RSI feature of the KFD is used to retrieve the KMF RSI programmed in the connected MR. The KMF RSI programmed in the MR identifies which KMF the MR is allowed to accept OTAR KMMs from. See section 3.8.9 for a detailed description of the procedures used for implementing this feature.

2.3.8 Load KMF RSI

The Load KMF RSI feature of the KFD is used to set the KMF RSI of a connected MR. See section 3.8.3 for a detailed description of the procedures used for implementing this feature.

2.3.9 View MNP

The View MNP feature of the KFD is used to retrieve the Message Number Period programmed in the connected MR. The MNP is used to determine if an OTAR KMM received via the RF channel is a replay of a previously sent KMM or if it is a newly sent KMM. See section 3.8.10 for a detailed description of the procedures used for implementing this feature.

2.3.10 Load MNP

The Load MNP feature of the KFD is used to set the MNP of a connected MR. See section 3.8.4 for a detailed description of the procedures used for implementing this feature.

2.3.11 View Keypad Info

The View Keypad Info feature of the KFD is used to retrieve information about the keysets being used in a connected MR. The keypad information to be viewed includes Keypad IDs, Keypad Names, and Keypad Format. See section 3.8.11 for a detailed description of the procedures used for implementing this feature.

2.3.12 Activate Keypad

The Activate Keypad feature of the KFD is used to activate a keypad in the connected MR. This feature is used in conjunction with crypto periods managed by the KMF. This feature works in conjunction with the View Keypad Info feature and requires that the View Keypad Info feature be implemented. See section 3.8.12 for a detailed description of the procedures used for implementing this feature.

2.4 Manual Rekeying Features for Authentication

2.4.1 Load Authentication Key

The Load Authentication Key feature of the KFD is used to transfer an authentication key from the KFD to a connected MR for storage with a SUID within the MR. See section 3.8.13 for a detailed description of the procedures used for implementing this feature.

2.4.2 Delete Authentication Key

The Delete Authentication Key feature of the KFD is used to erase the active SUID's authentication key or a specified SUID's authentication key or all authentication keys in a connected MR. See section 3.8.14 for a detailed description of the procedures used for implementing this feature.

2.4.3 View SUID Info

The View SUID Info feature of the KFD is used to view all the SUIDs in a connected MR. See section 3.8.15 for a detailed description of the procedures used for implementing this feature.

2.4.4 View Active SUID Info

The View Active SUID feature of the KFD is used to view the active SUID in a connected MR. See section 3.8.16 for a detailed description of the procedures used for implementing this feature.

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3 INTERFACE PROTOCOL DEFINITION

This section describes the protocols that are supported on the KFD-MR (Key Fill Device - Mobile Radio) interface. All protocols, from physical layer to application layer, are covered.

3.1 KFD-MR Interface

There are multiple possibilities for physical connection. It is required that interoperating KFDs and MRs implement the same one.

A protocol stack that can be supported on the KFD-MR interface using RS232 is shown in Figure 1. Only physical, link, network, transport and application layers are included on this interface. The details of the physical and link layers are described in sections 2.1.1, 2.1.2 and 2.2.2 of reference [2].

A protocol stack that can be supported on the KFD-MR interface using Three Wire Half Duplex Interface is shown in Figure 2. Only physical, link, and application layers are included on this interface. The physical, link and application layers of the Three Wire Half Duplex Interface are further described in Annex B of this document.

A protocol stack that can be supported on the KFD-MR interface using Universal Serial Bus (USB) is shown in Figure 3. Only physical, link, network, transport, and application layer protocols are included on this interface. The details of the physical and link layers are described in sections 2.1.2 and 2.2.2 of reference [2]. In addition, the Abstract Control Model (ACM) shall be used for the Communications Device Class (CDC) application layer (see section 3, Reference [9]). Using ACM, the USB device understands standard V.25ter (AT) commands and therefore provides backward-compatible RS232-over-USB support.

Manual Rekeying	Application
KFD-Interface Protocol	Application
UDP	Transport
IPv4	Network
PPP	Link
RS232	Physical

Figure 1. KFD-MR Protocol Stack (RS232)

Manual Rekeying	Application
3WI KFD-Interface Protocol	Application

4kbps Transfer Protocol	Link Physical
Three Wire Half Duplex Interface	

Figure 2. KFD-MR Protocol Stack (Three Wire Half Duplex Interface)

Manual Rekeying	Application
KFD-Interface Protocol	Application
UDP	Transport
IPv4	Network
PPP	Link
USB 2.0 CDC/ACM	Link
USB 2.0 Phy	Physical

Figure 3. KFD-MR Protocol Stack (USB)

Note: The Data Link Layer of Figure 3 consists of the USB 2.0 CDC/ACM and PPP protocols.

3.2 Physical Layer (RS232)

The physical interface between a KFD and compliant secure equipment loaded by a KFD consists of a serial RS232 connection. The complete description of the physical interface is found in section 2.1.1 of reference [2].

3.3 Physical Layer (USB)

The physical interface between a KFD and compliant secure equipment loaded by a KFD consists of a USB connection. The complete description of the physical interface is found in section 2.1.2 of reference [2].

3.4 Link Layer

A PPP connection is used at the Data Link Layer. See section 2.2.2 of Reference [2] for details.

3.5 Network Layer

IPv4 is used for communication. All IP addresses are in the unicast IP space; multicast and broadcast traffic is not used. Time-to-Live (TTL), Type of Service (TOS), and Precedence fields within IP headers are set to default values. See Reference [11] for additional details. The KFD learns the necessary IP addressing information from the MR during establishment of PPP connectivity.

3.6 Transport Layer

UDP is used for the transport layer (see Reference [10]). The MR expects to receive communications from a Key Fill Device on a configured UDP port, the default is 49165 as defined in TIA 102.BAJD Reference [7]. The KFD shall therefore use this as the destination port for messaging to the MR.

The source port for messages sent by the KFD and MR shall be selected by the MR and Key Fill Device. The valid range for the source port is 49165 – 65535. This may be the same as the port the MR expects to receive communications on.

The destination UDP port for messages sent by the Key Fill Device is established through configuration in the Key Fill Device prior to connection establishment. The supported range for the destination port is 49165 - 65535. The MR uses the source port in messaging received from the Key Fill Device as the destination port for messaging sent to the Key Fill Device. Minimally the destination port in the Key Fill Device needs to be configured to be the same as the listening port in the MR.

UDP checksums, as described in reference [10], shall be used for error detection.

3.7 KFD Interface Protocol Application Sublayer

The purpose of the KFD Interface Protocol is to provide a method for exchanging Key Management Messages (or KMMs) on the KFD interfaces. KMM Datagrams are used to deliver KMMs over the interface. The format of the KMM Datagram permits encryption. The encryption format for KMMs is defined in reference [1].

The KFD Interface Protocol also defines the procedure for the exchange of KMMs between the KFD and MR. This exchange procedure is referred to as an exchange session. During an exchange session, the KFD and MR can alternate roles of originator and recipient of KMM Datagrams. After a device sends a KMM Datagram on the interface, it shall go into receive to accept a KMM Datagram. Only then can it send a subsequent KMM Datagram.

3.7.1 Message Format

The Data Link Independent KMM datagram is used as the message format for the KFD interface protocol. When the KMM datagram is transmitted, the first field in each message body is transmitted first, the most significant octet in each field is transmitted first, and the most significant bit in each octet is transmitted first.

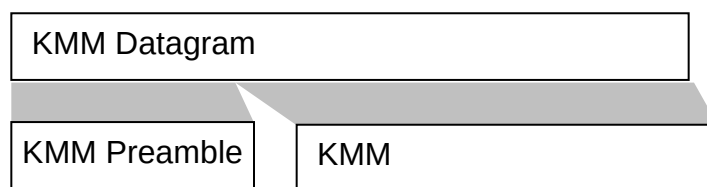


Figure 4. Data Link Independent KMM Datagram

The KMM Preamble and KMM are defined in reference [1].

3.7.2 Exchange Procedure

This section defines the exchange procedure, referred to as an exchange session that is used by the KFD Interface protocol. The protocol divides the exchange session into five parts. Refer to Figure 5.

- 1) Establish Session
- 2) KMM transfer from KFD to MR
- 3) KMM transfer from MR to KFD
- 4) Terminate Session
- 5) Disconnect

Step 1: The Establish Session handshake allows endpoint device type identification and shall be initiated by the KFD. The KFD sends a Session Control KMM containing the Session Control Opcode for Ready Request and Source Device Type for KFD. If the MR is capable of exchanging information with the KFD and is listening for a Session Control KMM, the MR responds with a Session Control KMM containing the Session Control Opcode for Ready General Mode and Source Device Type for MR, else it does not respond. If the KFD does not receive a response within 5 seconds, or if the KFD receives an invalid response, the KFD considers itself to be disconnected from the MR. If the proper response is received, the KFD considers itself to be connected, and the exchange session proceeds to step 2.

Note: Upon responding with a Session Control KMM, the MR should consider itself to be connected to the KFD until the KFD initiates logical disconnection (step 5) or as long as a general inactivity timer remains active. This MR timer is called the Key Fill Inactivity Timer (KIT). The length of the KIT may be configurable, its default value shall be 15 seconds, its minimum value shall be 5 seconds and its maximum value shall be 30 seconds. The KIT should be started in the MR upon sending a Session Control KMM to the KFD. The KIT is reset every time the MR sends a KMM to the KFD in subsequent steps. If its KIT

expires, the MR considers itself to be disconnected, and takes no further action, and will only respond to a subsequent Session Control KMM containing the Session Control Opcode for Ready Request and Source Device Type for KFD.

Step 2: The KFD now transfers any and all KMMs queued for this MR. To minimize the memory requirements of the MR, the KFD transfers the KMMs one at a time waiting for a reply after each transfer. The MR replies with a KMM datagram containing the appropriate response KMM. Once all KMMs have been transferred, the KFD sends a Session Control KMM containing the Session Control Opcode for Transfer Done and the Source Device Type for KFD indicating to the MR that it has completed its transfer. The exchange session now moves to step 3.

Note: If the KFD does not receive a valid KMM from the MR within 5 seconds, the KFD terminates the session (see step 4) and may attempt to initiate another session in the manner described in step 1.

Step 3: The MR now transfers any and all KMMs queued for the KFD in the same manner as step 2. The MR transfers the KMMs one at a time waiting for a reply after each transfer. The KFD replies with a KMM datagram containing the appropriate response KMM. Once all KMMs have been transferred, or if there are no KMMs to transfer, the MR sends a Session Control KMM containing the Session Control Opcode for Transfer Done and Source Device Type for MR to indicate to the KFD that it has completed its transfer. The exchange session now moves to step 4.

Step 4: The KFD now terminates the session by sending a Session Control KMM containing the Session Control Opcode for End Session and Source Device Type for KFD to the MR. In response, the MR sends a Session Control KMM containing the Session Control Opcode for End Session Ack and Source Device Type for MR, and the session is terminated. If the KFD does not receive a response within 5 seconds, or if the KFD receives an invalid response, the KFD terminates the session. Upon termination of the session, the KFD takes one of the following actions: a) initiates a new session by moving to Step 1; b) initiates logical disconnection of the MR by moving to Step 5.

Step 5: The KFD initiates the disconnection after completing the session termination from Step 4-b. This is shown in Figure 5. The KFD sends a Session Control KMM containing the Session Control Opcode for Disconnect and Source Device Type for KFD to the MR. In response, the MR sends a Session Control KMM containing the Session Control Opcode for Disconnect Ack and a Source Device Type for MR, and the MR then cancels the KIT and considers itself to be disconnected from the KFD. If the KFD does not receive a response within 5 seconds, or if the KFD receives an invalid response, the KFD considers itself to be logically disconnected from the MR and no further action is taken.

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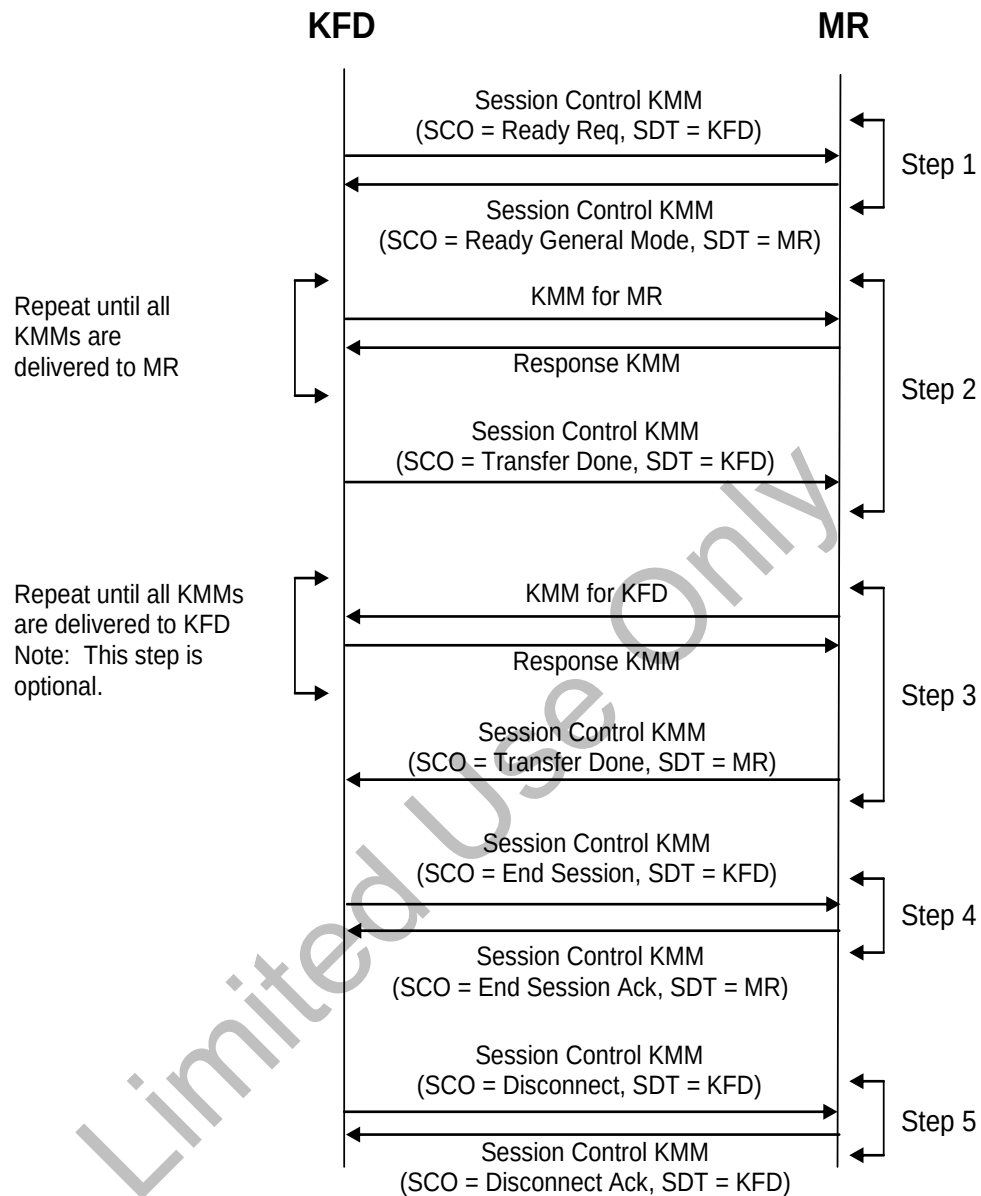


Figure 5. Exchange Session defined by the KFD- Interface Protocol

While physically connected to the MR, a KFD can initiate multiple exchange sessions. Each exchange session is preceded by a Ready Req Session Control KMM and consists of the five steps described above.

The KFD or MR shall accept and use a received KMM regardless of the success or failure of delivery for subsequent KMMs within the exchange session.

A KFD may also execute an exchange session without actually transferring any KMMs. This would be done for the purpose of determining if the connected MR

is capable of exchanging information using the KFD Interface Protocol. Such an exchange session would consist of the following KMM datagrams being exchanged:

1. Session Control KMM (Ready Req) from KFD to MR
2. Session Control KMM (Ready General Mode) from MR to KFD
3. Session Control KMM (Transfer Done) from KFD to MR
4. Session Control KMM (Transfer Done) from MR to KFD
5. Session Control KMM (End Session) from KFD to MR
6. Session Control KMM (End Session Ack) from MR to KFD
7. Session Control KMM (Disconnect) from KFD to MR
8. Session Control KMM (Disconnect Ack) from MR to KFD

From this exchange session, the KFD could determine that a MR was connected.

3.8 Manual Rekeying Application Sublayer

The Manual Rekeying application provides the manual rekeying KFD features. The Manual Rekeying application determines what KMMs need to be transferred for the particular KFD feature selected. The KMMs are then passed to KFD Interface Protocol for transfer to the MR. This is described in Step 2 in Section 3.7.2.

The following sections identify the manual rekeying procedures that are used to implement the manual rekeying features. A manual rekeying procedure consists of a KMM being sent from the KFD to the MR and the appropriate KMM response being sent from the MR to the KFD. For some manual rekeying features, a series of manual rekeying procedures is required.

The MR shall respond with a Negative Acknowledgement KMM (Status = \$03 Invalid Message ID) for all messages that the MR does not understand.

Furthermore, KMMs sent between the KFD and MR may be optionally outer-layer encrypted. The Algorithm ID, Key ID and Message Indicator used for the outer-layer encryption of the KMM are indicated in the KMM Preamble, defined in Ref [1]. If the MR cannot decrypt an outer-layer encrypted KMM sent by the KFD, it shall respond with the Unable to Decrypt Response KMM.

3.8.1 Keyload

To load encryption keys, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List Active Keyset IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
2. The KFD sends one or more Modify-Key-Command KMMs to the MR. The MR replies to each with a Rekey-Ack KMM.

The complete exchange session showing all KMMs for the Keyload feature is shown below.

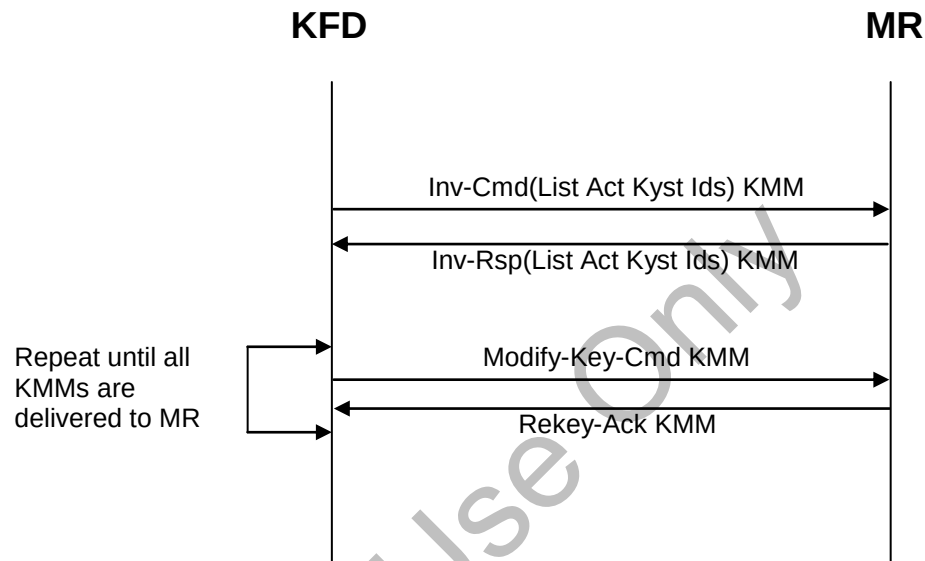


Figure 6. Exchange Session for the Keyload feature

The keyload procedure supports the loading of multiple keys. Multiple keys of the same algorithm can be contained in a single Modify Key Command KMM. The size of the KMM is restricted to a predetermined length (see section 3.9). The KFD repeats the Modify Key Command exchange until all keys are transferred.

Special mention is given to how keysets are initialized in the MR. The necessary operations are described as follows:

1. The MR has no keysets enabled or activated when first brought into service. An Inventory query issued by the KFD results in an indication of no keysets being found.
2. The first keyset of any order within a crypto group that is populated with keys from the KFD is automatically enabled and activated.
3. If a new keyset in a crypto group is populated with keys, and there already is an active and enabled keyset in the crypto group, the new keyset is enabled but not activated. The currently active keyset remains active until an Activate Keyset procedure changes its activation state. There is no scenario where the only enabled keyset in a crypto group is not active.
4. New keysets are enabled by use of the Modify Key command.

3.8.2 Load Individual RSI

To load an individual RSI into a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List RSI Items) KMM to the MR. The MR replies with an Inventory-Response KMM.
2. The KFD sends a Change RSI Command KMM to the MR. The MR replies with a Change RSI Response KMM.

The complete exchange session showing all KMMs for the Load the Individual RSI feature is shown below.

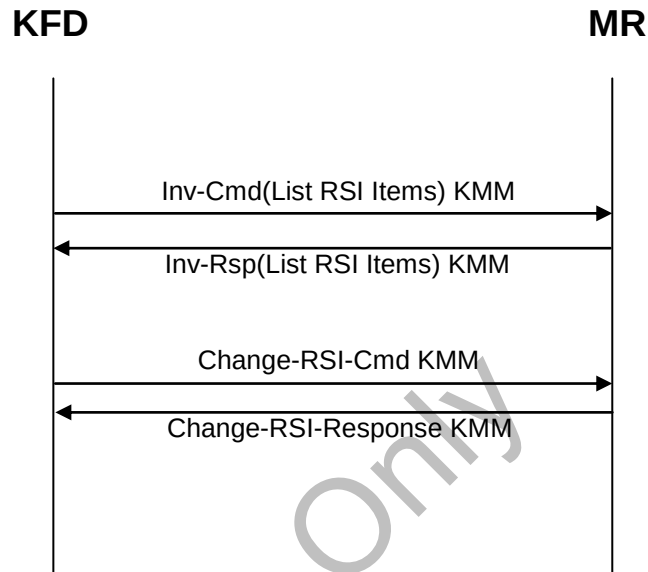


Figure 7. Exchange Session for the Load Individual RSI feature

The Message Number field within the KFD version of Change RSI Command KMM shall always be set to zero. No other values are allowed, nor should they be accepted by the MR. The MR shall correspondingly set its inbound and outbound Message Numbers to zero upon receiving the Change RSI Command KMM. This requirement applies to any RSI, Individual or Group, to which a Change RSI Command originating from the KFD was targeted.

3.8.3 Load KMF RSI

To program the proper the KMF RSI into a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List MNP) KMM to the MR. The MR replies with an Inventory-Response KMM.

2. The KFD sends a Load-Config Command KMM with the MNP from the MR and the new KMF RSI. The MR replies with a Load-Config-Response KMM.

The complete exchange session showing all KMMs for the Load KMF RSI feature is shown below.

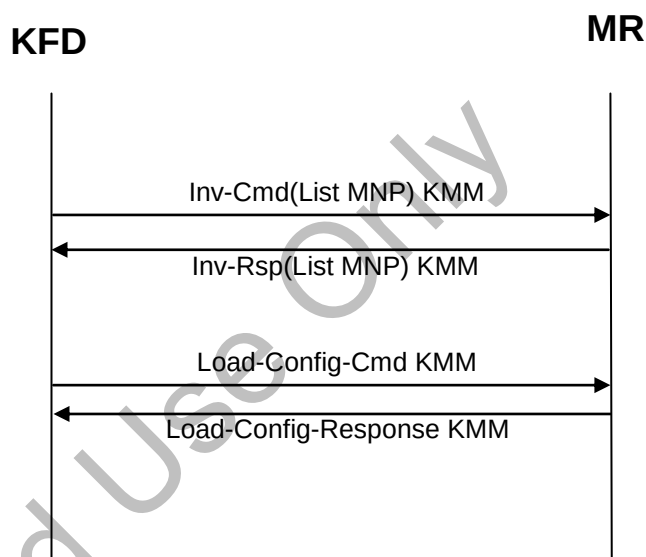


Figure 8. Exchange Session for the Load KMF RSI feature

3.8.4 Load MNP

To program the proper Message Number Period into a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List KMF RSI) KMM to the MR. The MR replies with an Inventory-Response KMM.

2. The KFD sends a Load-Config Command KMM with the KMF RSI from the MR and the new MNP. The MR replies with a Load-Config-Response KMM.

The complete exchange session showing all KMMs for the Load MNP feature is shown below.

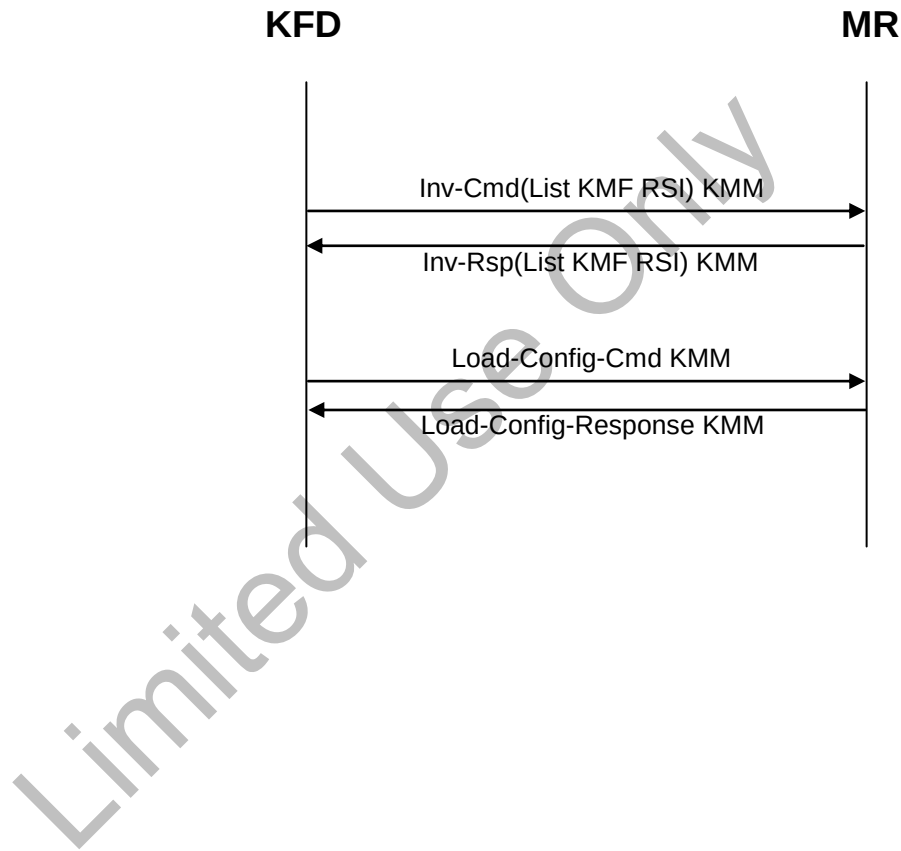


Figure 9. Exchange Session for the Load MNP feature

3.8.5 Erase Keys

To erase keys in a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List Active Keyset IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
2. The KFD sends a Modify-Key-Command KMM identifying the key to be erased. Only the key in the specified SLN and keyset is erased.

The Delete/Rekey bar of the Key Format primitive field is set to %1 in this case. The MR replies to each with a Rekey-Ack KMM.

The complete exchange session showing all KMMs for the Erase Keys feature is shown below.

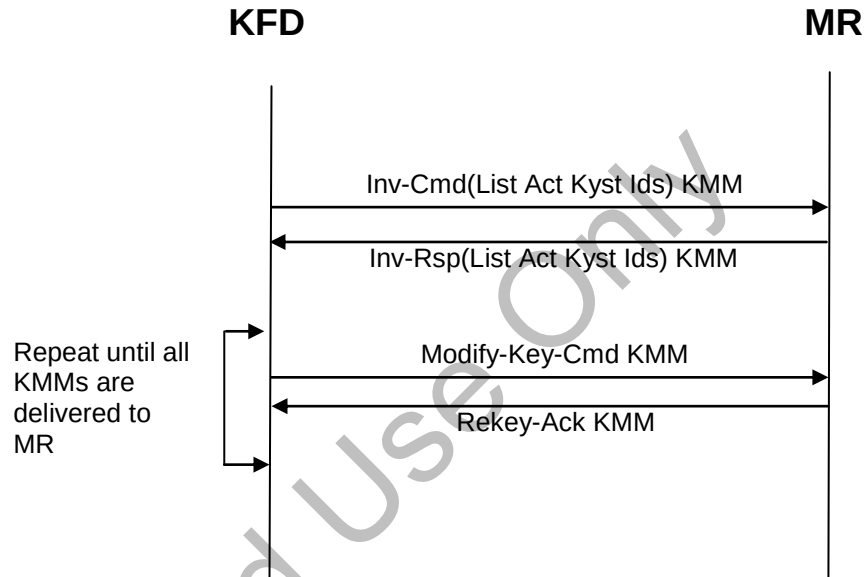


Figure 10. Exchange Session for the Erase Keys feature

The key erase procedure supports the deletion of multiple keys. The deletion commands for multiple keys can be contained in a single Modify Key Command KMM. The size of the KMM is restricted to a predetermined length (see section 3.9). The KFD repeats the Modify Key Command exchange until all key deletion notifications are transferred.

3.8.6 Erase All Keys

To erase all keys in a connected MR, the following KMM exchange is recommended:

1. The KFD sends a Zeroize-Command KMM to the MR. The MR replies with a Zeroize-Response KMM.

The complete exchange session showing all KMMs for the Erase All Keys feature is shown below.

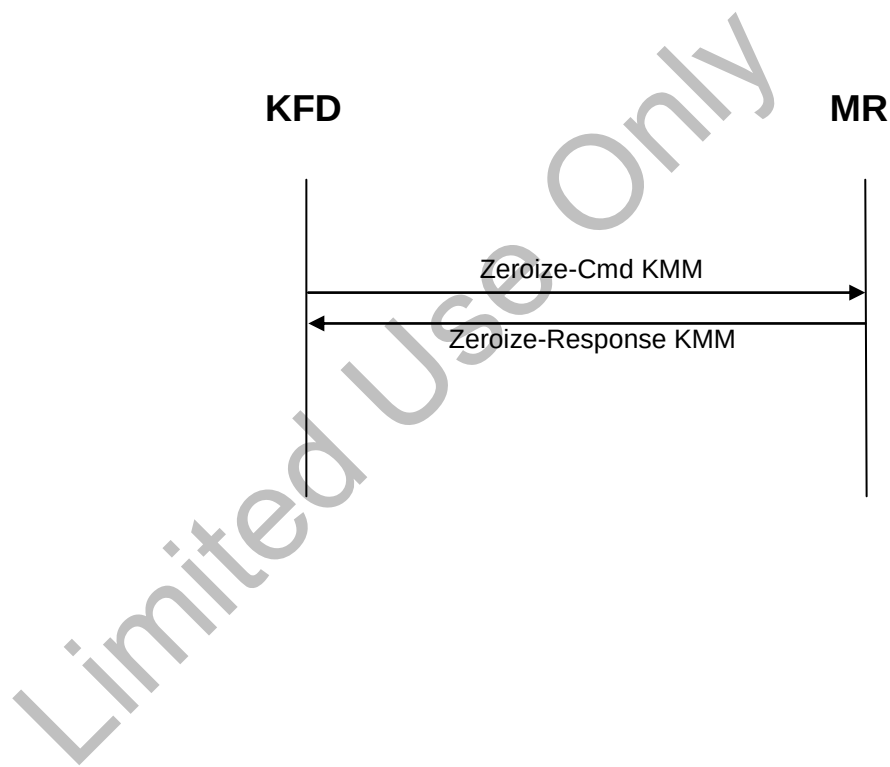


Figure 11. Exchange Session for the Erase All Keys feature

3.8.7 View Key Info

To view information about the keys contained in a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List Active Keys) KMM to the MR. The MR replies with an Inventory-Response KMM.
2. Repeat step 1 if the number of keys exceeds what can be returned in a single KMM.

The complete exchange session showing all KMMs for the View Key Info feature is shown below.

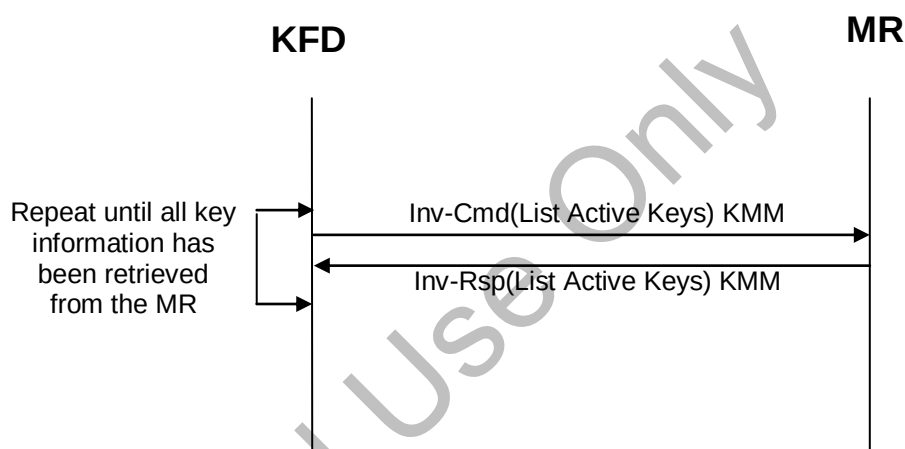


Figure 12. Exchange Session for the View Key Info feature

3.8.8 View Individual RSI

To view the individual RSI of the connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List RSI Items) KMM to the MR. The MR replies with an Inventory-Response KMM.

Note: The first RSI returned shall be the individual RSI.

The complete exchange session showing all KMMs for the View Individual RSI feature is shown below.

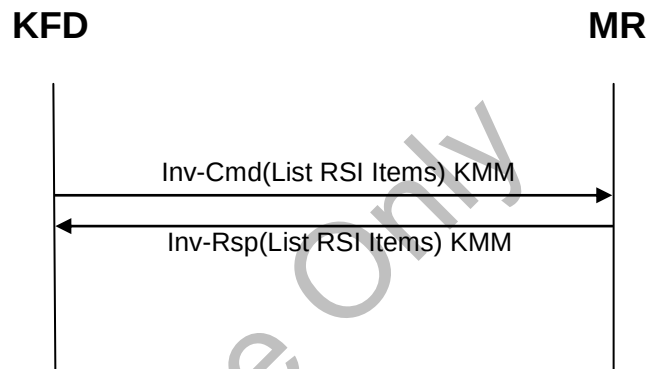


Figure 13. Exchange Session for the View Individual RSI feature

3.8.9 View KMF RSI

To view the KMF RSI programmed in a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List KMF RSI) KMM to the MR. The MR replies with an Inventory-Response KMM.

The complete exchange session showing all KMMs for the View KMF RSI feature is shown below.

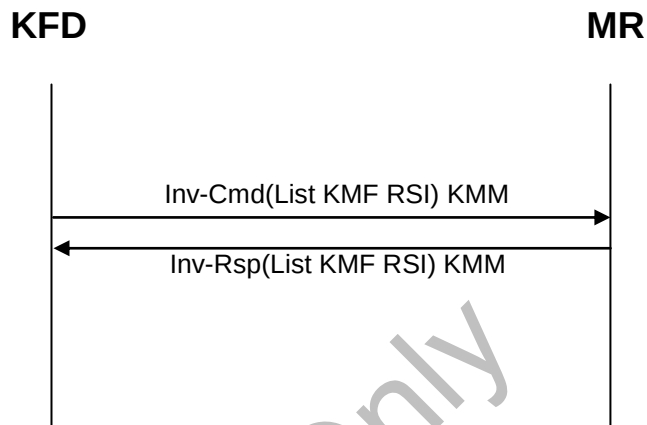


Figure 14. Exchange Session for the View KMF RSI feature

3.8.10 View MNP

To view the Message Number Period programmed in the connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List MNP) KMM to the MR.
The MR replies with an Inventory-Response KMM.

The complete exchange session showing all KMMs for the View MNP feature is shown below.

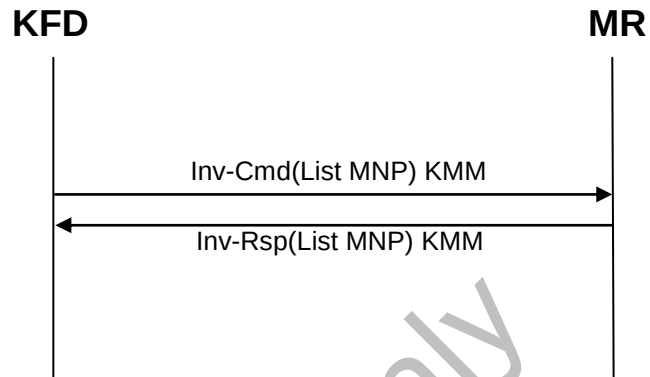


Figure 15. Exchange Session for the View MNP feature

3.8.11 View Keyset Info

To view keyset information stored in a connected MR, the following KMM exchange is recommended:

1. The KFD sends an Inventory-Command (List Active Keyset IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
2. The KFD sends an Inventory-Command (List Keyset Tagging Info) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. Repeat step 2 View Keyset Info if the keyset information exceeds what can be returned in a single KMM.

The complete exchange session showing all KMMs for the View Keyset Info feature is shown below.

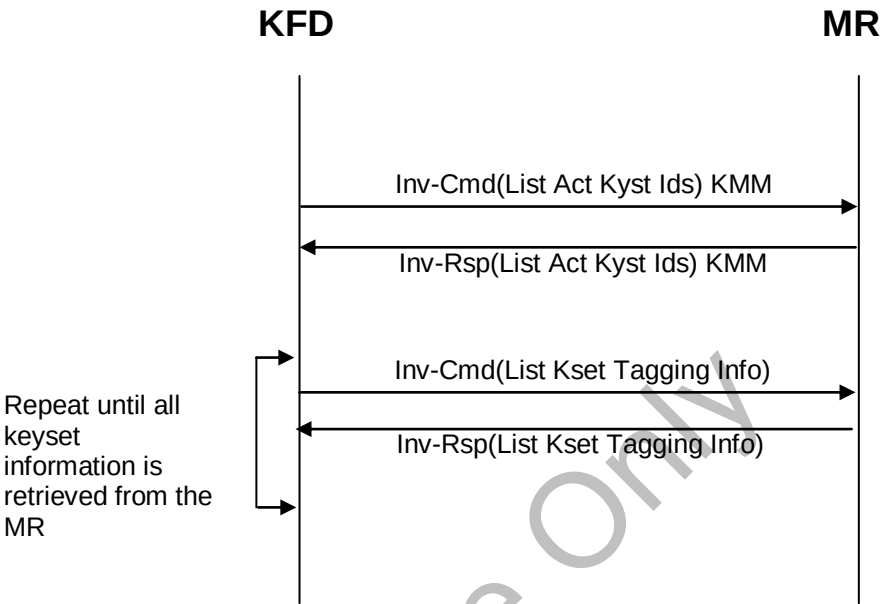


Figure 16. Exchange Session for the View Keyset Info feature

3.8.12 Activate Keyset

To activate a keyset in the connected MR, the following KMM exchange is recommended:

1. The KFD sends a Changeover-Command KMM to the MR. The MR replies with a Changeover-Response KMM.

The complete exchange session showing all KMMs for the Activate Keyset feature is shown below.

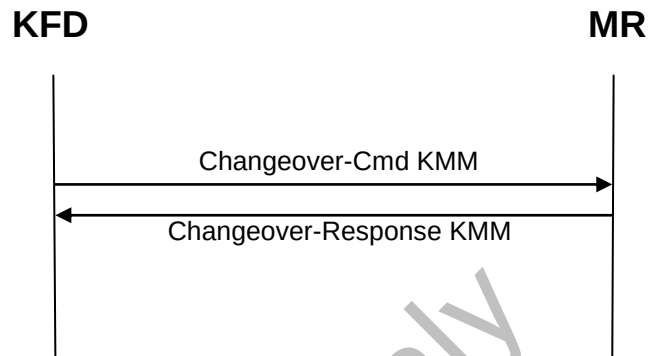


Figure 17. Exchange Session for the Activate Keyset feature

3.8.13 Load Authentication Key

To load an Authentication key into the connected MR, the following exchange is initiated:

1. The KFD sends a Load Authentication Key -Command KMM to the MR, and the MR replies with a Load Authentication Key-Response KMM containing the SUID that was loaded with the authentication key.

The complete exchange session showing all KMMs for the Load Authentication Key feature is shown below. Each run of the complete exchange session can load one authentication key. If multiple authentication keys need to be loaded, then the complete exchange session is run multiple times.

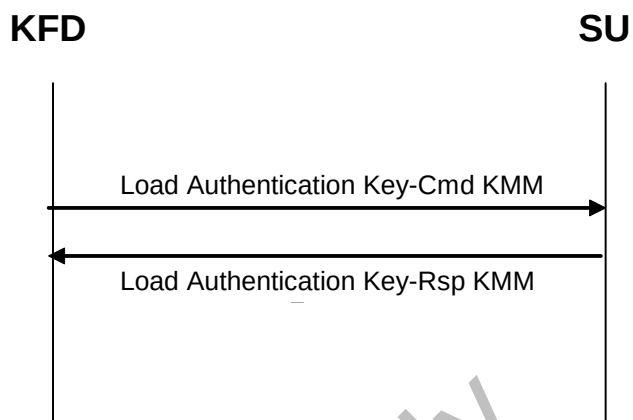


Figure 18. Exchange Session for the Load Authentication Key feature

3.8.14 Delete Authentication Key

To delete an authentication key from the connected MR, the following exchange is initiated:

1. The KFD sends a Delete Authentication Key -Command KMM to the MR and the MR replies with a Delete-Authentication-Key-Response KMM.

The complete exchange session showing all KMMs for the Delete Authentication Key feature is shown below. If multiple authentication keys need to be deleted, then the complete exchange session is run multiple times.

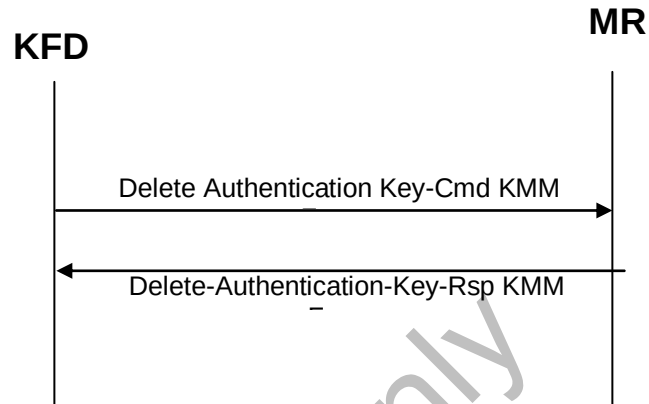


Figure 19. Exchange Session for the Delete Authentication Key feature

3.8.15 View SUID Info

To view information about the SUID contained in a connected MR, the following exchange is performed:

1. The KFD sends an Inventory-Command (List SUID Items) KMM to the MR and the MR replies with an Inventory-Response KMM containing the SUIDs in the MR. (This step is repeated for a number of SUIDs that exceeds what can be returned in a single KMM).

The complete exchange session showing all KMMs for the View SUID Info feature is shown below.

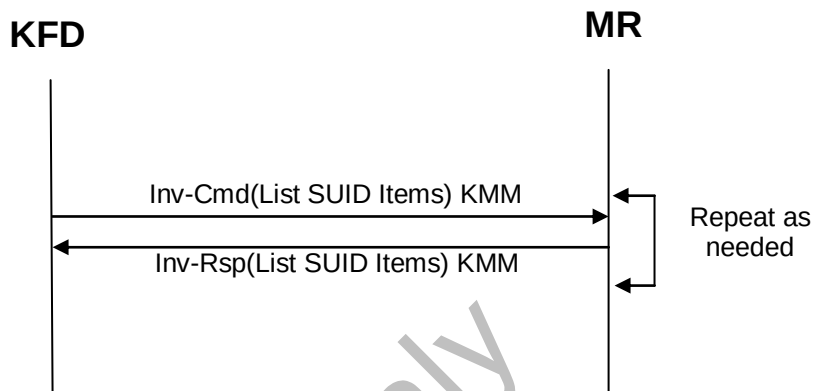


Figure 20. Exchange Session for the View SUID Info feature

3.8.16 View Active SUID Info

The Active SUID is the one that the MR uses in its current operating mode when transacting trunking services with the RFSS, and with other MRs. To view information about the Active SUID in a connected MR and optionally load an Authentication key, the following exchange is performed:

1. The KFD sends an Inventory-Command (List Active SUID) KMM to the MR and the MR replies with an Inventory-Response KMM containing the Active SUID in the MR.
2. (Optional Step) The KFD sends a Load Authentication Key -Command KMM to the MR, and the MR replies with a Load Authentication Key-Response KMM containing the SUID that was loaded with the authentication key.

The complete exchange session showing all KMMs for the View SUID Info feature is shown below. Each run of the complete exchange session may load only one authentication key. If multiple authentication keys need to be loaded, then the complete exchange session is run multiple times.

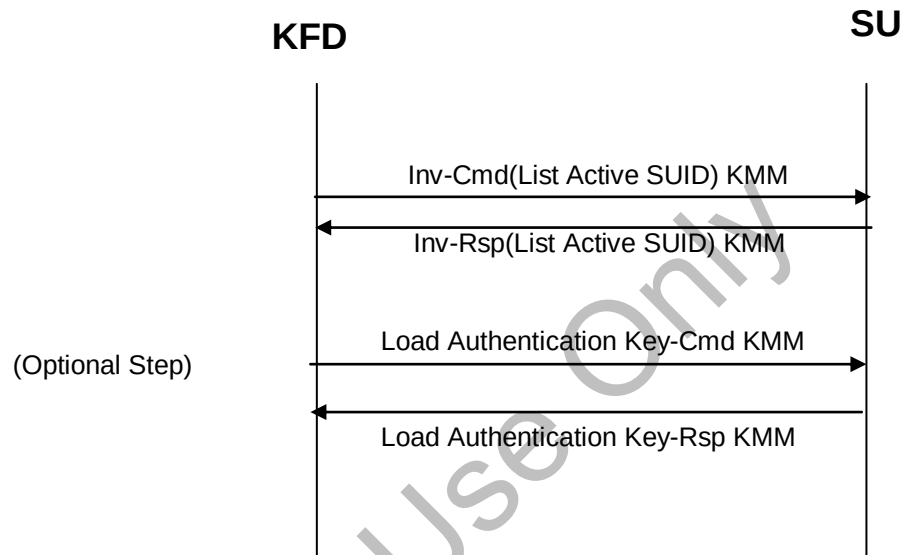


Figure 21. Exchange Session for the View Active SUID Info feature

3.9 KMMs

The KFD Interface Protocol uses KMMs defined in Table 1. The KFD interface uses several KMMs that are used by OTAR (see [1]). Detailed definitions of the fields within the KMMs are defined in the Primitive Field Definitions of reference [1]. In addition, the KFD interface may require the ability to transfer additional information that is not required over the air. In these cases, a KFD Interface Protocol KMM has been defined to replace the normal KMM (for example, Modify-Key-Command). The rightmost column of Table 1 specifies whether or not each KMM is reused from OTAR.

The size of the data link-independent KMM shall not exceed 478 octets.

Keyloading KMM	Msg ID:Inv Type	Valid Rsp Kind	Re-used OTAR KMM
Change-RSI-Command	\$03	3	yes
Change-RSI-Response	\$04	1	yes
Changeover-Command	\$05	3	yes
Changeover-Response	\$06	1	yes
Inventory-Cmd (List Active Kset IDs)	\$0D:\$02	3	yes
Inventory-Rsp (List Active Kset IDs)	\$0E:\$02	1	yes
Inventory-Cmd (List RSI Items)	\$0D:\$0B	3	yes
Inventory-Rsp (List RSI Items)	\$0E:\$0B	1	yes
Inventory-Rsp (List Active SUID)	\$0E:\$F7	1	no
Inventory-Rsp (List SUID Items)	\$0E:\$F8	1	no
Inventory-Cmd (List Active SUID)	\$0D:\$F7	3	no
Inventory-Cmd (List SUID Items)	\$0D:\$F8	3	no
Inventory-Cmd (List Kset Tagging Info)	\$0D:\$F9	3	no
Inventory-Rsp (List Kset Tagging Info)	\$0E:\$F9	1	no
Inventory-Cmd (List Active Keys)	\$0D:\$FD	3	no
Inventory-Rsp (List Active Keys)	\$0E:\$FD	1	no
Inventory-Cmd (List MNP)	\$0D:\$FE	3	no
Inventory-Rsp (List MNP)	\$0E:\$FE	1	no
Inventory-Cmd (List KMF RSI)	\$0D:\$FF	3	no
Inventory-Rsp (List KMF RSI)	\$0E:\$FF	1	no
Modify-Key-Command	\$13	3	no
Negative-Ack	\$16	1	yes
Rekey-Acknowledgment	\$1D	1	yes
Zeroize-Command	\$21	3	yes
Zeroize-Response	\$22	1	yes
Unable-to-Decrypt-Response	\$27	1	yes
Load Authentication Key-Command	\$28	3	no
Load Authentication Key-Response	\$29	1	no
Delete Authentication Key-Command	\$2A	3	no
Delete Authentication Key-Response	\$2B	1	no
Session Control	\$31	1	no
Load-Config-Response	\$FC	1	no
Load-Config-Command	\$FD	3	no

Table 1. Manual Rekeying KMM List

Note: Unable to Decrypt Response KMM is not supported on the Three-Wire Half-Duplex Interface, as defined in Annex B.

3.9.1 KMM Header Definition

The KMM header for all KMMs defined in this document is as defined in Reference [1]. The KMM header shall be filled as follows:

- Message ID field set appropriately
- Message Length field set appropriately

- Rsp field in Message Format set appropriately
- MN field in Message Format always set to 0 (No Message Number Used)
- MAC field in Message Format always set to 0 (No MAC Used)
- The KFD shall use \$FFFFFFF as the Destination RSI if it doesn't know the Individual RSI value of the MR; the MR shall always accept a KMM whose Destination RSI is \$FFFFFFF. Otherwise, the KFD may use the Individual RSI of the target MR as the Destination RSI, or may continue to use \$FFFFFFF as the Destination RSI. The MR shall set the Destination RSI to the KFD's Individual RSI value for KMMs returned from the MR to the KFD.
- The Source RSI field is set appropriately. The MR is permitted to accept any Source RSI, including the All-Call (\$FFFFFFF).

3.9.2 KMM Body Definitions

The KMM body formats are defined below.

3.9.2.1 Change-RSI-Command

See Reference [1].

3.9.2.2 Change-RSI-Response

See Reference [1].

3.9.2.3 Changeover-Command

See Reference [1].

3.9.2.4 Changeover-Response

See Reference [1].

3.9.2.5 Inventory-Command (List Active Kset IDs)

See Reference [1].

3.9.2.6 Inventory-Response (List Active Kset IDs)

See Reference [1].

3.9.2.7 Inventory-Command (List RSI Items)

See Reference [1].

3.9.2.8 Inventory-Response (List RSI Items)

See Reference [1].

3.9.2.9 Inventory-Command (List Keyset Tagging Info)

Octet 0	Inventory Type
---------	----------------

Table 2. Inventory-Command (List Keyset Tagging Info) Message Format

3.9.2.10 Inventory-Response (List Keyset Tagging Info)

The Inventory-Response (List Keyset Tagging Info) used on the KFD Interface Protocol is the same as defined in Reference [1] except for the value of the Inventory Type field.

3.9.2.11 Inventory-Command (List Active Keys)

Octet 0	Inventory Type
1 - 3	Inventory Marker
4 - 5	Max # of Keys Requested

Table 3. Inventory-Command (List Active Keys) Message Format

Inventory Marker – An arbitrary value that can be used by the target to mark the position in the list of keys. It is used when all keys cannot be returned in a single KMM because the size of the KMM would exceed the maximum size allowed. As noted in section 3.9, the maximum size of a KMM is 478 octets. A value of 0 in the Inventory Marker field of the Inventory-Command indicates the target should start at the beginning of its internal list of keys. A non-zero value identifies a position in the list of keys other than the beginning.

Note: The value of the Inventory Marker is arbitrary and can differ for different radio manufacturers. The meaning of the value is only known to the target radio.

Max # of keys requested – Identifies the max number of keys that should be returned in a single KMM. Used by the KFD to limit the size of the KMM returned by the target.

Typically, the first Inventory-Command (List Active Keys) sets the Inventory Marker field to 0 and the Max # of Keys Requested field to 78. The Inventory Marker field set to zero indicates that the target should start at the beginning of its list of keys. The Max # of Keys Requested field limits the Inventory-Response (List Active Keys) to 478 octets by allowing only 78 Key Info Blocks to be returned. If the target contains more than 78 keys, then it should set the Inventory Marker field in the Inventory-Response KMM to a value that marks the 79th position in the list of keys. If the KFD wishes to retrieve the additional key info blocks, it should send a second Inventory-Command (List Active Keys) with

the Inventory Marker field set to the value received in the previous Inventory-Response. This indicates to the target that it should start with the key in the 79th position rather than the key in the 1st position. When the target sends an Inventory-Response containing the remainder of the keys, it shall set the Inventory Marker to zero. In this way, the KFD can retrieve all key info blocks stored in the target without exceeding the maximum size of a KMM.

3.9.2.12 Inventory-Response (List Active Keys)

Octet 0	Inventory Type	
1 - 3	Inventory Marker	
4 - 5	Number of Items	Number of Items = x
6	Keyset ID	SEQUENCE [1..x] of
7 - 8	SLN	Key Info Items
9	Algorithm ID	
10 - 11	Key ID	
12	Subsequent Key Info Items	

Table 4. Inventory-Response (List Active Keys) Message Format

Inventory Marker – See section 3.9.2.11 for a description of how this field is used. Set to 0 to indicate no additional keys. Set to an arbitrary value marking the position of the next key if all keys were not sent in the Inventory-Response.

3.9.2.13 Inventory-Command (List MNP)

Octet 0	Inventory Type
---------	----------------

Table 5. Inventory-Command (List MNP) Message Format

3.9.2.14 Inventory-Response (List MNP)

Octet 0	Inventory Type
1 - 2	Message Number Period

Table 6. Inventory-Response (List MNP) Message Format

3.9.2.15 Inventory-Command (List KMF RSI)

Octet 0	Inventory Type
---------	----------------

Table 7. Inventory-Command (List KMF RSI) Message Format**3.9.2.16 Inventory-Response (List KMF RSI)**

Octet 0	Inventory Type
1 - 3	KMF RSI

Table 8. Inventory-Response (List KMF RSI) Message Format**3.9.2.17 Modify-Key-Command**

The Modify-Key-Command used on the KFD Interface Protocol has been modified to add one additional octet. The fields shown in Table 11 are defined in Reference [1] except for the Extended Decryption Instruction Format octet and the Key field when the key is sent unencrypted.

Extended Decryption Instruction octet

The Extended Decryption Instruction Format octet is defined as follows:

Octet 1	0	0	0	0	0	0	0
	7	6	5	4	3	2	1

The most significant bit (b7) is unavailable and shall be set to zero. The other bits (b6 – b0) are reserved and should be set to zero. The reserved bits may be defined in the future.

Key field (unencrypted key transfer)

The key transmitted in the Modify Key Command is typically sent unencrypted. If the key is to be encrypted, it shall be done in accordance with Reference [1].

For DES, as described in Reference [1], the plaintext key bits shall be placed in the Key field as follows:

Octet 0	Key Bits B ₁ ..B ₇ , P ₁
1	Key Bits B ₈ ..B ₁₄ , P ₂
2	Key Bits B ₁₅ ..P ₂₁ , P ₃
3	Key Bits B ₂₂ ..B ₂₈ , P ₄
4	Key Bits B ₂₉ ..B ₃₅ , P ₅
5	Key Bits B ₃₆ ..B ₄₂ , P ₆
6	Key Bits B ₄₃ ..B ₄₉ , P ₇
7	Key Bits B ₅₀ ..B ₅₆ , P ₈
	7 6 5 4 3 2 1 0

Key Field Format

Octet 0 - This octet contains the left most (most significant) 7-bits of the plaintext key and the first parity bit. The left most (most significant) key bit, B_1 , is placed in the left most (most significant) position of the octet and the parity bit is placed in the right most (least significant) position of the octet.

Octet 1 - This octet contains the next 7 bits of the plaintext key and the second parity bit.

...

Octet 7 - This octet contains the last (least significant) 7 bits of the plaintext key and the last parity bit.

Table 9. DES Key Format

For algorithms that use block encryption protocol (TDES and AES), as described in Reference [1], the plaintext key shall be transmitted without the check vector. The plaintext key bits shall be placed in the Key field as follows:

Octet 0	Key bits $b_1..b_8$
1	Key bits $b_9..b_{16}$
$k/8-1$	Key bits $b_{(k/8*8)+1}..b_k$
	7 6 5 4 3 2 1 0

Key Field Format

Octet 0 - This octet contains the left most (most significant) 8 bits of the plaintext key. The left most (most significant) key bit, b_1 , is placed in the left most (most significant) position of the octet.

Octet 1 - This octet contains the next 8 bits of the plaintext key.

...

Octet $k/8-1$ - This octet contains the remaining bits of the plaintext key. If the remaining key bits do not fill the octet, the key bits shall be left justified and the remaining unused bits shall be filled with zeros.

Table 10. Block Encryption Protocol Key Field Format

Octet	0	Decryption Instruction Format	Decryption Instruction Block
	1	Extended Decryption Instr Fmt	
	2	Algorithm ID	
	3	Key ID	
	4		
	5	Update Count (Optional) [0..1]	
5 or 6	6	Message Indicator (Optional) [0 or 9 octets]	
(i = 0,1,9,10)	i+3		End of Block
	i+4	Keyset ID	
	i+5	Algorithm ID	
	i+6	Key Length	
	i+7	Number of Keys	
	i+8	Key Format	
	i+9	Storage Location Number	Number of Keys = x SEQUENCE [1..x] of Unique Key Items
	i+10		
	i+11	Key ID	
	i+12		
	i+13	Key [Key Length octets]	
i+13+Key Length		Checksum (Optional) [0 or 4 octets]	
i+14+Key Length			
		Key Name (Optional) [0..31 octets]	
		Subsequent Unique Key Items (if they exist)	
variable			
		7 6 5 4 3 2 1 0	

Table 11. Modify-Key-Command Message Format

3.9.2.18 Negative-Acknowledgment

See Reference [1].

3.9.2.19 Rekey-Acknowledgment

See Reference [1].

3.9.2.20 Zeroize-Command

See Reference [1].

3.9.2.21 Zeroize-Response

See Reference [1].

3.9.2.22 Load-Config-Command

Octet	0 - 2	KMF RSI
	3 - 4	Message Number Period

Table 12. Load-Config-Command Message Format

3.9.2.23 Load-Config-Response

Octet	0 - 2	KMF RSI
	3 - 4	Message Number Period
	5	Status

Table 13. Load-Config-Response Message Format

Status – A value of 0 means KMF RSI and MNP were accepted. A value of 1 means the Command Was Not Performed.

3.9.2.24 Unable to Decrypt Response

See Reference [1]. When the Unable to Decrypt Response KMM is used on the Key Fill Interface, it shall not be outer-layer encrypted. Furthermore, the following optional blocks shall not be used: Decryption Instruction, Reverse Warm Start and Unique Key Item.

3.9.2.25 Load Authentication Key – Command

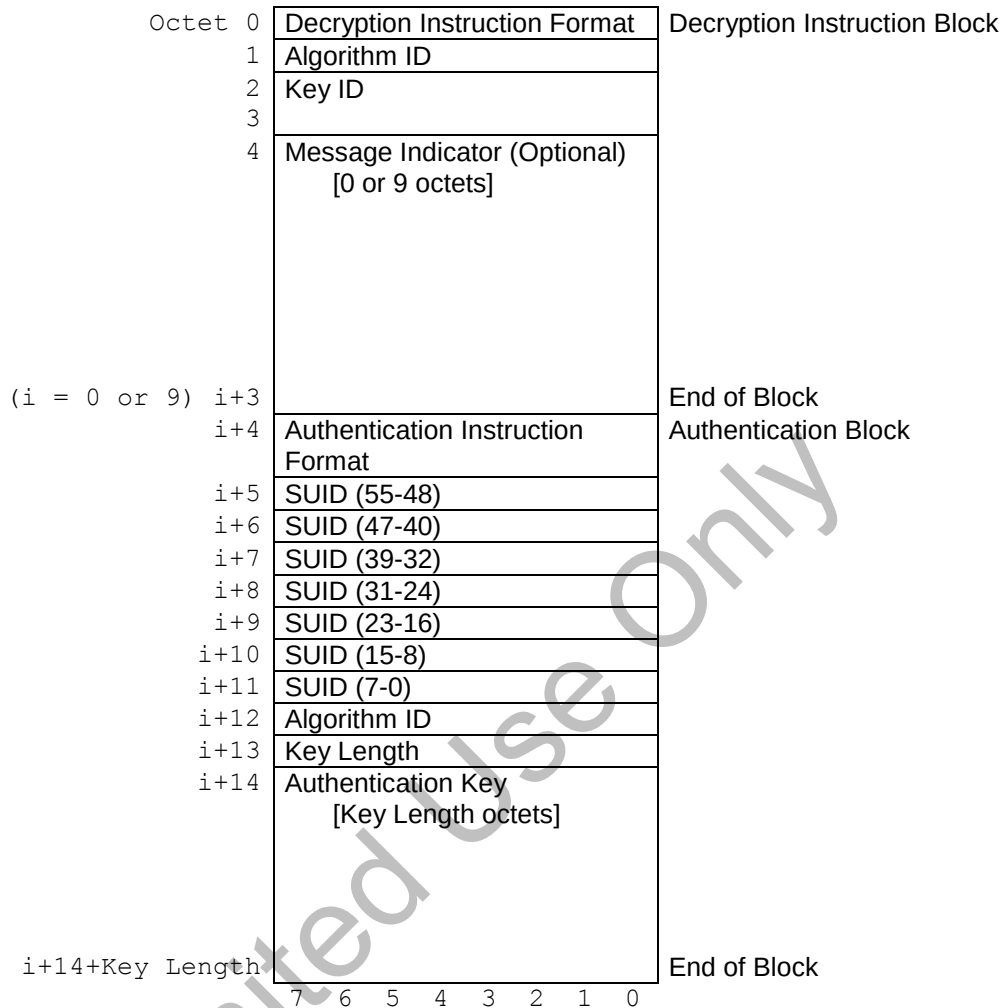


Table 14. Load Authentication Key-Command Message Format

Decryption Instruction Block

Decryption Instruction Format - Indicates which optional fields are included in the Decryption Instructions Block field. The format for this field is defined in the Primitive Field Definition section for Decryption Instruction Format.

Algorithm ID - The Algorithm ID is used in conjunction with the Key ID to uniquely select a KEK. The format for this field is defined in the Primitive Field Definition section for Algorithm ID. Algorithm ID = \$80 Unencrypted value.

Key ID - The Key ID of the KEK. – when used. The format for this field is defined in the Primitive Field Definition section for Key ID.

Message Indicator (Optional) - Provides the Message Indicator (encryption synchronization) for encryption algorithms which do not support the Electronic Codebook mode of operation. The format for this field is defined in the Primitive Field Definition section for Message Indicator.

Authentication Instruction Format

B0 = SUID null or defined

A value of 0 indicates the Active SUID is targeted by the command; a value of 1 indicates the SUID specified by the SUID field is targeted by the command. B1 = Is unused and set to zero

B2 = Is unused and set to zero

B3 = Is unused and set to zero

B4 = Is unused and set to zero

B5 = Is unused and set to zero

B6 = Is unused and set to zero

B7 = Is unused and set to zero

Algorithm ID – The Algorithm ID identifies the algorithm used with the Authentication Key. The format for this field is defined in the Primitive Field Definition section for Algorithm ID. Algorithm ID = \$85 AES-128

Key Length - The number of octets used to transfer the key. The format for this field is defined in the Primitive Field Definitions section for Number.

SUID (55-48) – If assigned = WACN ID (19-12)

SUID (47-40) – If assigned = WACN ID (11-4)

SUID (39-36) – If assigned = WACN ID (3-0)

SUID (35-32) – If assigned = System ID (11-8)

SUID (31-24) – If assigned = System ID (7-0)

SUID (23-16) – If assigned = Subscriber ID (23-16)

SUID (15-8) – If assigned = Subscriber ID (15-8)

SUID (7-0) – If assigned = Subscriber ID (7-0)

If the SUID is not specified, then the SUID fields are 0.

The authentication key is made immediately active once it is successfully received by the MR in the Load Authentication Key-Command message. There is no changeover procedure for authentication keys.

For the AES algorithm the plaintext key bits shall be placed in the Key field as follows:

Octet 0	Key bits $b_1..b_8$
1	Key bits $b_9..b_{16}$
$k/8-1$	Key bits $b_{(k/8+1)..b_k}$
	7 6 5 4 3 2 1 0

Table 15. Key Field Format

Key Field Format

Octet 0 - This octet contains the left most (most significant) 8 bits of the plaintext key. The left most (most significant) key bit, b_1 , is placed in the left most (most significant) position of the octet.

Octet 1 - This octet contains the next 8 bits of the plaintext key.

...

Octet $k/8-1$ - This octet contains the remaining bits of the plaintext key

3.9.2.26 Load Authentication key – Response

Octet 0	Authentication Instruction Format
1	SUID (55-48)
2	SUID (47-40)
3	SUID (39-32)
4	SUID (31-24)
5	SUID (23-16)
6	SUID (15-8)
7	SUID (7-0)
8	Status
	7 6 5 4 3 2 1 0

Table 16. Load Authentication Key-Response Message Format

Authentication Instruction Format

B0 = failed/successful SUID assignment to authentication key, (K) A value of 0 indicates failed assignment of SUID to K. A value of 1 indicates successful assignment of SUID to K.

B1 = Is unused and set to zero

B2 = Is unused and set to zero

B3 = Is unused and set to zero

B4 = Is unused and set to zero

B5 = Is unused and set to zero

B6 = Is unused and set to zero

B7 = Is unused and set to zero

SUID (55-48) – WACN ID (19-12)

SUID (47-40) – WACN ID (11-4)

SUID (39-36) – WACN ID (3-0)

SUID (35-32) – System ID (11-8)

SUID (31-24) – System ID (7-0)

SUID (23-16) – Subscriber ID (23-16)

SUID (15-8) – Subscriber ID (15-8)

SUID (7-0) – Subscriber ID (7-0)

If the SUID is not assigned, then the SUID fields are 0. The Status field reflects the error status; the SUID is not always zeroed for all errors.

Status

Status field values are defined in Reference [1] and shown in Table 17

Error precedence, from highest 1) to lowest 4)

- 1) \$0E Invalid WACN or System ID
 - 2) \$0F Invalid Subscriber ID
 - 3) Any of
 - \$05 Out of memory
 - \$09 Invalid Algorithm ID
 - \$0B Module Failure
 - 4) \$01 Command could not be performed (unspecified reason)
- Note that any error status found in Table 17 that is not mentioned above is not relevant to the key fill operation for Link Layer Authentication.

Status	Reason	Value (Hex)	Values (Binary)
Command was performed	Entire command was successfully performed	\$00	0000 0000
Command could not be performed (unspecified reason)	Command could not be performed due to a reason other than specified in the Status field	\$01	0000 0001
Item (Key/Keyset) does not exist	Key / Keyset to perform operation on does not exist	\$02	0000 0010
Invalid Message ID	Message ID is invalid	\$03	0000 0011
Invalid Checksum / MAC	MAC or Checksum is invalid	\$04	0000 0100
Out of Memory	Memory unavailable to process the command / message	\$05	0000 0101
Could not decrypt the message (KEK did not exist)	KEK did not exist to decrypt the message	\$06	0000 0110
Invalid Message Number	Message Number is invalid	\$07	0000 0111
Invalid Key ID	Key ID is not present	\$08	0000 1000
Invalid Algorithm ID	Algorithm ID is not valid or present	\$09	0000 1001
Invalid MFID	MFID is not valid	\$0A	0000 1010
Module Failure	Encryption Hardware failure	\$0B	0000 1011
MI all zeros	Received MI was all zeros	\$0C	0000 1100
Keyfail	Key identified by ALGID/KEYID is erased	\$0D	0000 1101
Invalid WACN or System ID	WACN ID or System ID not present	\$0E	0000 1110
Invalid Subscriber ID	Subscriber ID not valid or present	\$0F	0000 1111
Reserved for Future Use	Reserved	\$10 - \$FE	0001 0000 – 1111 1110
Unknown	Unknown	\$FF	1111 1111

Table 17. Status Field Values**3.9.2.27 Delete Authentication Key – Command**

Octet 0	Authentication Instruction Format
1	SUID (55-48)
2	SUID (47-40)
3	SUID (39-32)
4	SUID (31-24)
5	SUID (23-16)
6	SUID (15-8)
7	SUID (7-0)
	7 6 5 4 3 2 1 0

Table 18. Delete Authentication Key-Command Message Format*Authentication Instruction Format*

B0 = SUID unassigned/assigned to an authentication key, (K) A value of 0 indicates the active SUID is targeted by the command; a value of 1 indicates the SUID specified by the SUID field is targeted by the command.

B1 = Erase all or currently active

Valid if B0 = 0, A value of 1 for B1 indicates to erase all authentication keys. A value of 0 indicates to erase the currently active SUID's authentication key.

If B0 = 1, then B1 is unused and set to zero.

B2 = Is unused and set to zero

B3 = Is unused and set to zero

B4 = Is unused and set to zero

B5 = Is unused and set to zero

B6 = Is unused and set to zero

B7 = Is unused and set to zero

SUID (55-48) – WACN ID (19-12)

SUID (47-40) – WACN ID (11-4)

SUID (39-36) – WACN ID (3-0)

SUID (35-32) – System ID (11-8)

SUID (31-24) – System ID (7-0)

SUID (23-16) – Subscriber ID (23-16)

SUID (15-8) – Subscriber ID (15-8)

SUID (7-0) – Subscriber ID (7-0)

If the SUID is not assigned, then the SUID fields are 0.

3.9.2.28 Delete Authentication Key-Response

Octet	0	SUID (55-48)
	1	SUID (47-40)
	2	SUID (39-32)
	3	SUID (31-24)
	4	SUID (23-16)
	5	SUID (15-8)
	6	SUID (7-0)
	7	Number of Keys Deleted
	8	
	9	Status
		7 6 5 4 3 2 1 0

Table 19. Delete Authentication Key-Response Message Format

SUID (55-48) – WACN ID (19-12)

SUID (47-40) – WACN ID (11-4)

SUID (39-36) – WACN ID (3-0)

SUID (35-32) – System ID (11-8)

SUID (31-24) – System ID (7-0)

SUID (23-16) – Subscriber ID (23-16)

SUID (15-8) – Subscriber ID (15-8)

SUID (7-0) – Subscriber ID (7-0)

The Active SUID is the one that the MR uses in its current operating mode when transacting trunking services with the RFSS, and with other MRs. For erase all authentication keys the currently active SUID is specified. For erase all or currently active, if there was no currently active SUID, then SUID fields are 0. The specified SUID is located in the SUID fields.

Number of Keys Deleted

0 = No keys deleted.

1- 65535 = Number of keys deleted

If a SUID does not have a key it is not counted. Erase all authentication keys, status command was performed and no keys deleted indicates all SUID in the MR did not have keys before the Delete Key Command was received.

Status

Status field values are defined in Reference [1] and shown in Table 17. If the SUID(s) did not have a key before the command was received the Status value will be \$01 Command could not be performed. If delete currently active and there is no active SUID then status indicates an invalid SUID by stating \$0E Invalid WACN or System ID.

Error precedence, from highest 1) to lowest 4)

- 1) \$0E Invalid WACN or System ID
- 2) \$0F Invalid Subscriber ID
- 3) \$0B Module Failure
- 4) \$01 Command could not be performed (unspecified reason)

Note that any error status found in Table 17 that is not mentioned above is not relevant to the key fill operation for Link Layer Authentication.

3.9.2.29 Inventory-Command (List SUID Items)

Octet 0	Inventory Type
1 -3	Inventory Marker
4 -5	Max # of SUID Requested
	7 6 5 4 3 2 1 0

Table 20. Inventory-Command (List SUID Items) Message Format

Inventory Type - The Inventory Type specifies which items need to be sent in the Inventory Response message. The format for this field is defined in the Primitive Field Definitions section for Inventory Type.

Inventory Marker – An arbitrary value that can be used by the target to mark the position in the list of SUID. It is used when all SUID cannot be returned in a single KMM because the size of the KMM would exceed the maximum size allowed. As noted in section 3.9, the maximum size of a KMM is 478 octets. A value of 0 in the Inventory Marker field of the Inventory-Command indicates the target should start at the beginning of its internal list of SUID. A non-zero value identifies a position in the list of SUID other than the beginning.

Note: The value of the Inventory Marker is arbitrary and can differ for different radio manufacturers. The meaning of the value is only known to the target radio.

Max # of SUID requested – Identifies the max number of SUID that should be returned in a single KMM. Used by the KFD to limit the size of the KMM returned by the target.

Typically, the first Inventory-Command (List SUID Items) sets the Inventory Marker field to 0 and the Max # of SUID Requested field to 59. The Inventory Marker field set to zero indicates that the target should start at the beginning of its list of SUID. The Max # of SUID Requested field limits the Inventory-Response (List SUID Items) to 478 octets by allowing only 59 SUID K Status to be returned. If the target contains more than 59 SUID, then it should set the Inventory Marker field in the Inventory-Response KMM to a value that marks the 60th position in the list of SUID. If the KFD wishes to retrieve the additional SUID K Status, it should send a second Inventory-Command (List SUID Items) with the Inventory Marker field set to the value received in the previous Inventory-Response. This indicates to the target that it should start with the key in the 60th position rather than the key in the 1st position. When the target sends an Inventory-Response containing the remainder of the SUID, it shall set the Inventory Marker to zero. In this way, the KFD can retrieve all key info blocks stored in the target without exceeding the maximum size of a KMM.

3.9.2.30 Inventory-Response (List SUID Items)

Octet 0	Inventory Type	SEQUENCE [1..n] of SUID and K Status
1 - 3	Inventory Marker	
4	Number of Items	
5	SUID (55-48)	
6	SUID (47-40)	
7	SUID (39-32)	
8	SUID (31-24)	
9	SUID (23-16)	
10	SUID (15-8)	
11	SUID (7-0)	
12	K status	
4+n*8	Subsequent SUID and K Status (if they exist)	
	7 6 5 4 3 2 1 0	

Table 21. Inventory-Response (List SUID Items) Message Format

Inventory Marker – See section 3.9.2.11 for a description of how this field is used. Set to 0 to indicate no additional SUID. Set to an arbitrary value marking the position of the next key if all SUID were not sent in the Inventory-Response.

SUID (55-48) – WACN ID (19-12)

SUID (47-40) – WACN ID (11-4)

SUID (39-36) – WACN ID (3-0)

SUID (35-32) – System ID (11-8)

SUID (31-24) – System ID (7-0)

SUID (23-16) – Subscriber ID (23-16)

SUID (15-8) – Subscriber ID (15-8)

SUID (7-0) – Subscriber ID (7-0)

SUID K Status

B0 = SUID unassigned/assigned to an authentication key, (K) A value of 0 indicates no K assigned. A value of 1 indicates K is assigned.

B1 = Currently active SUID

A value of 0 indicates it is not the currently active SUID. A value of 1 indicates it is the currently active SUID.

B2 = Is unused and set to zero

B3 = Is unused and set to zero

B4 = Is unused and set to zero

B5 = Is unused and set to zero

B6 = Is unused and set to zero

B7 = Is unused and set to zero

3.9.2.31 Inventory-Command (List Active SUID)

Octet 0	Inventory Type
	7 6 5 4 3 2 1 0

Table 22. Inventory-Command (List Active SUID) Message Format

Inventory Type - The Inventory Type specifies which items need to be sent in the Inventory Response message. The format for this field is defined in the Primitive Field Definitions section for Inventory Type.

3.9.2.32 Inventory-Response (List Active SUID)

Octet 0	Inventory Type
1	Inventory Instruction Format
2	SUID (55-48)
3	SUID (47-40)
4	SUID (39-32)
5	SUID (31-24)
6	SUID (23-16)
7	SUID (15-8)
8	SUID (7-0)
9	Status
	7 6 5 4 3 2 1 0

Table 23. Inventory Response (List Active SUID) Message Format

Inventory Type - The Inventory Type specifies which items need to be sent in the Inventory Response message. The format for this field is defined in the Primitive Field Definitions section for Inventory Type.

Inventory Instruction Format

B0 = SUID field is null or specified

A value of 0 indicates no SUID. A value of 1 indicates a SUID is specified; this is the currently active SUID in the SU.

B1 = K assigned to currently active SUID

A value of 0 indicates no K is assigned. A value of 1 indicates a K is assigned.

If B0 = 0, then B1 is unused and set to zero.

B2 = Is unused and set to zero

B3 = Is unused and set to zero

B4 = Is unused and set to zero

B5 = Is unused and set to zero

B6 = Is unused and set to zero

B7 = Is unused and set to zero

SUID (55-48) – WACN ID (19-12)
 SUID (47-40) – WACN ID (11-4)
 SUID (39-36) – WACN ID (3-0)
 SUID (35-32) – System ID (11-8)
 SUID (31-24) – System ID (7-0)
 SUID (23-16) – Subscriber ID (23-16)
 SUID (15-8) – Subscriber ID (15-8)
 SUID (7-0) – Subscriber ID (7-0)

If the SUID is null, then the SUID fields are 0.

Status

Status field values are defined in Reference [1] and shown in Table 17.

If there is no active SUID then status indicates an invalid SUID by stating \$0E Invalid WACN or System ID.

Error precedence, from highest 1) to lowest 3)

- 1) \$0E Invalid WACN or System ID
- 2) \$0B Module Failure
- 3) \$01 Command could not be performed (unspecified reason)

Note that any error status found in Table 17 that is not mentioned above is not relevant to the key fill operation for Link Layer Authentication.

3.9.2.33 Session Control KMM

Octet 0: Version

Octet 0 of the Session Control KMM body contains a version number. The formats of the octets that follow are defined by the Version.

Session Control Message Body – Version = \$00:

Octet 0	Version
1	Session Control Opcode
2	Source Device Type
	7 6 5 4 3 2 1 0

Table 24. Session Control Message Format

Octet 1 - Session Control Opcode

\$00: not used

\$01: Ready Request

\$02: Ready General Mode

\$03: Transfer Done

\$04: End Session

\$05: End Session Ack

\$06: Disconnect

\$07: Disconnect Ack

\$08-\$FF: Reserved

Octet 2 – Source Device Type

The Source Device Type identifies the device type of the originator of a transmitted Session Control KMM. The Source Device Type is set based upon which device is sending the Session Control KMM.

\$00: not used

\$01: KFD

\$02: MR

\$03-\$FF: Reserved

Limited Use Only

Annex A (Normative): CRC Generation (Register & State Table)

The 16-bit CRC uses the CRC-16-CCITT generator polynomial: $G(X) = X^{16} + X^{12} + X^5 + 1$. The Intermediate CRC (CRCi) is then calculated as:

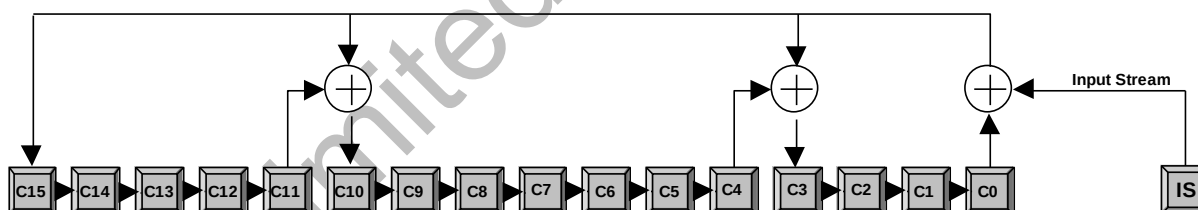
$$CRCi = \frac{data_stream}{G(X)}$$

Finally, CRC is equal to CRCi with the most significant and least significant bytes swapped.

Figure 22 shows a method of realizing this CRC calculator using a shift register. To generate a CRC, initialize [C15:C0] to \$FFFF and start shifting in the data for which a CRC shall be calculated.

The following figure shows the various states of the shift register while calculating the CRC for the sample data stream \$DEAD. Each 8-bit word in the data stream is shifted into the register with the least significant bit being sent first.

Figure 22. Shift Register Depiction of the CRC Generation



In this figure, [C15:C0] represents the CRC register. After all data bits for which a CRC needs to be calculated are shifted in, [C15:C0] represents the CRCi for that data stream. [IS] represents the next bit of the data stream to be shifted into the register.

Figure 23. State Table for CRC Generator Shift Register

Shift #	C15	C14	C13	C12	C11	C10	C9	C8	C7	C6	C5	C4	C3	C2	C1	C0		IS
Initial	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		0
1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1	1		1
2	0	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1		1
3	0	0	1	1	1	1	1	0	1	1	1	1	1	1	0	1		1
4	0	0	0	1	1	1	1	1	0	1	1	1	1	1	1	0		1
5	1	0	0	0	1	0	1	1	1	0	1	1	0	1	1	1		0
6	1	1	0	0	0	0	0	1	1	1	0	1	0	0	1	1		1
7	0	1	1	0	0	0	0	0	1	1	1	0	1	0	0	1		1
8	0	0	1	1	0	0	0	0	0	1	1	1	0	1	0	0		1
9	1	0	0	1	1	1	0	0	0	0	1	1	0	0	1	0		0
10	0	1	0	0	1	1	1	0	0	0	0	1	1	0	0	1		1
11	0	0	1	0	0	1	1	1	0	0	0	0	1	1	0	0		1
12	1	0	0	1	0	1	1	1	1	0	0	0	1	1	1	0		0
13	0	1	0	0	1	0	1	1	1	1	0	0	0	1	1	1		1
14	0	0	1	0	0	1	0	1	1	1	1	0	0	0	1	1		0
15	1	0	0	1	0	1	1	0	1	1	1	1	1	0	0	1		1
16	0	1	0	0	1	0	1	1	0	1	1	1	1	1	0	0		X

After all of the bits in the data stream have been shifted into the register, the last state of the CRC register is the CRC_i for the data stream. CRC is completed by swapping the least significant and most significant bytes of CRC_i. In this example, CRC_i for the sample data stream \$DEAD is \$4B7C and CRC is \$7C4B.

Limited Use Only

Annex B (Normative): Three Wire Half Duplex Interface

B.1 Physical Layer (Three Wire Half Duplex Interface)

The three-wire half duplex interface is defined for backwards compatibility. The physical interface between a KFD and any secure capable equipment loaded by a KFD consists of three wires: K/F, Keyload*, and GND.

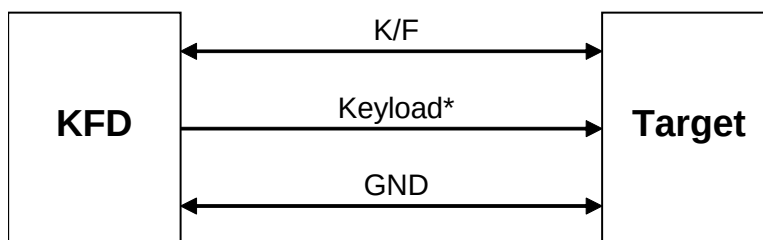


Figure 24. KFD Physical Interface Lines (Three Wire Half Duplex Interface)

B.1.1 K/F Line Description

The K/F line is a bi-directional communications line used to transfer data to or from a MR device at 4kbps. Since only a single wire is used, only half-duplex operation is possible. This is the primary communications line.

With the K/F line configured for receive:

- $V_{ILmax} = 1.3$ volts.
- $V_{IHmin} = 3.7$ volts.

With the K/F line configured for transmit, the KFD circuit consists of a BJT transistor with a 10k pullup to 5 volts:

- KFD transmit low: $V_{OLmax} = 0.5$ volts.
- KFD transmit high: depends on load circuit. The KFD has a 10k pullup to 5 volts.

Tolerance on 4kbps baud rate:

- Baud rate transmitted by KFD = 4kbps +/- 1%
- Baud rate received by KFD = 4kbps +/- 3%. The KFD can handle up to 3% error on the baud rate and still receive correctly.

B.1.1.1 Sending a Key Signature

Once the KFD is physically connected to the MR, the KFD operator can initiate an exchange session. When an exchange session is initiated, the KFD sends a 100mS low pulse, referred to as a key signature, on the K/F line. The key signature synchronizes the MR to the start of a new exchange session (See section B.2.7.3). After releasing the line (bringing it high), the KFD pauses for

5mS. Once this pause is complete, the KFD is ready to exchange data on the interface.

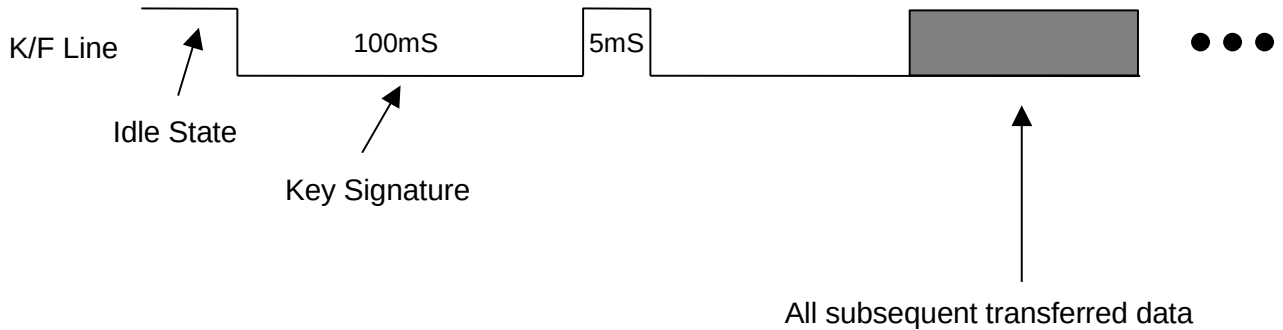


Figure 25. Sending a Key Signature (Three Wire Half Duplex Interface)

B.1.2 Keyload* Line Description

The Keyload* line is shorted to GND in the KFD. It may be used by mobile radios to sense a KFD has been connected to the MR. This allows the MR to operate in a lower power state and monitor the K/F line only after the KFD has been attached.

Note: The “*” in Keyload* simply denotes that the line is active low.

B.1.3 GND Line Description

The GND line provides a common reference between the KFD circuits and the MR device’s circuits.

B.2 Link Layer (4kbps Transfer Protocol) (Three Wire Half Duplex Interface)

Once a key signature is sent on the K/F line as described in section B.1.1.1 the KFD and target can exchange data using the 4kbps transfer protocol described in this section. The 4kbps transfer protocol provides the following:

1. It defines the format and timing for the transfer of a single byte of data.
2. It provides a mechanism for pacing the data transfer to assure data is not lost.
3. It provides a mechanism for turning around the direction of data flow on the K/F line.
4. It defines a mechanism for detection of a lost connection through the use of timers.

B.2.1 4kbps Transfer Protocol Byte Format

Figure 26 shows the format for the transfer of a single byte of data on the K/F line. The transfer starts with the K/F line in the IDLE (or high) state. The sender then drives the line low for 250uS. This is referred to as a Start Bit. The 8 bits of data, D7 thru D0, are then sent for 250uS each, starting with the most significant bit, D7, and ending with the least significant bit, D0. Next, an even parity bit is sent for 250uS. Finally, the K/F line is driven low for at least 1 mS to complete the transmission.

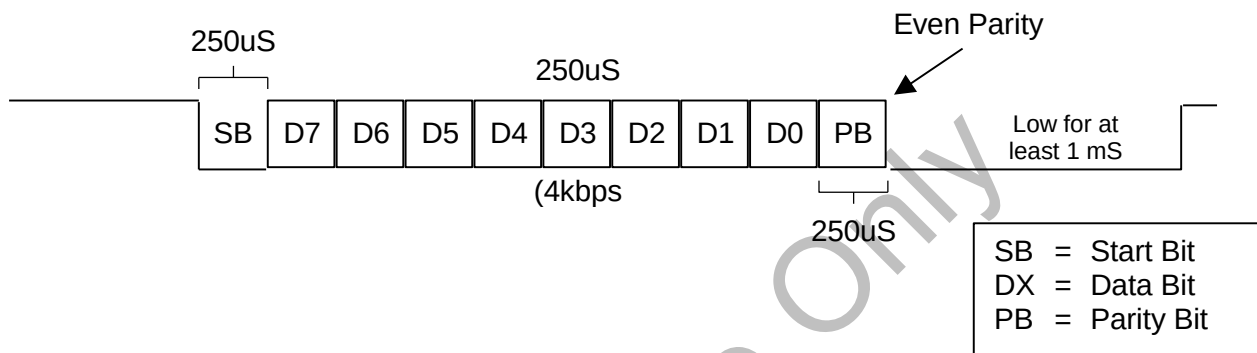


Figure 26. Anatomy of a Byte (Three Wire Half Duplex Interface)

B.2.2 4kbps Transfer Protocol Flow Control

To accomplish flow control, two elementary states are defined, BUSY & IDLE. A logical high on the K/F line while data is not being transferred indicates that the interface is in the IDLE state. A logical low on the K/F line while data is not being transferred indicates that the interface is in the BUSY state. These elements provide for a simple method of flow control as follows.

A device wishing to send a byte shall check the K/F line and only send the byte if the line is IDLE. As long as the K/F line is BUSY, the sender shall not transmit. Once the byte has been transmitted as described in section B.2, the receiver holds the K/F line BUSY until it is ready to receive the next byte. The receiver then releases the K/F line, allowing it to go IDLE. This simple method allows a device to send a frame consisting of many bytes while assuring that the receiver is ready to receive each of those bytes.

B.2.3 4kbps Transfer Protocol Line Turnaround

Since only a single line, K/F, is used to carry data, only half duplex operation is possible. The KFD interface protocol is setup so the direction of data flow alternates. Back to back frames are never sent in the same direction. After a frame is sent from the KFD to a connected MR, the next frame is from the

connected MR to the KFD, and so on. The 4kbps transfer protocol provides a mechanism to allow the direction of data flow on the K/F line to be turned around as follows.

Upon sending the last byte of a frame, the K/F line should be held BUSY by the sender while it reconfigures its K/F line port for input operation. When the device is ready to receive, it releases the K/F line, allowing it to go IDLE. A receiving device, upon receiving the last byte of a message, reconfigures its K/F line for transmit operation.

The 4kbps transfer protocol operates on bytes. The KFD interface protocol shall indicate to the 4kbps transfer protocol where the end of a frame is so the K/F line can be turned around. On the sending end, as bytes are provided to the 4kbps transfer protocol for transmit on the K/F line, an 'end of frame' indication shall be provided. On the receiving end, as bytes are received, they are passed up to the KFD interface protocol. Once an entire frame is received, an 'end of frame' indication shall be provided to the 4kbps transfer protocol so the line can be turned around.

B.2.4 Timeout Considerations

The 4kbps transfer protocol provides a mechanism for detecting that a connected device is not responding. Two timers are utilized for this purpose, t_{out1} and t_{out2} . t_{out1} is used to detect that a receiver is not present or not ready. t_{out1} is started when a device is ready to send but the K/F line is BUSY. The sender shall wait for the K/F line to go IDLE. If the K/F line does not go IDLE within the defined time, a timeout error has occurred.

t_{out2} is used to detect that a sender is not present or not ready. t_{out2} is started when a device is ready to receive. If a start bit is not detected within the defined time, a timeout error has occurred.

Once a timeout occurs, the communicating device flags an error and no further data should be exchanged. A target device should return to waiting for a new key signature. A KFD should start subsequent communications with a new key signature.

KFD Version	t_{out1}	t_{out2}
Mode	5 Sec	5 Sec

Figure 27. Timeout Values (Three Wire Half Duplex Interface)

B.2.5 Transmitting a Byte (TXBYTE)

The following procedure can be used by a device wishing to send a byte of data. It takes into account flow control and line turnaround requirements.

- Start t_{out1}
- Release K/F line to ensure that it's not being driven by the sender
- Wait for K/F line to go IDLE (stop t_{out1} when K/F goes IDLE)
- Transmit Start Bit (1 @ 250uS/bit)
- Transmit Data bits (8 @ 250uS/bit)
- Transmit Parity bit (1 @ 250uS/bit - Even)
- Hold K/F Line BUSY For 1mS
- Terminate TXBYTE leaving K/F line BUSY.

Note: The K/F line is left BUSY upon termination. This prevents the receiver from turning the line around and sending immediately if this is the last byte in a frame.

B.2.6 Receiving a Byte (RXBYTE)

The following procedure can be used by a device expecting a byte of data. It takes into account flow control and line turnaround requirements.

- Start t_{out2}
- Release K/F line to ensure that it's not being driven by the receiver.
- Wait for K/F line to go IDLE
- Wait for falling edge (Start bit) on K/F line (stop t_{out2} when Start bit is detected)
- Sample trailing data bits (8 @ 250uS/bit)
- Sample parity bit (1 @ 250uS/bit - Even)
- Check for correct parity.
- Terminate RXBYTE leaving K/F line BUSY.

Note: The K/F line is left BUSY upon termination to keep the sender from sending another byte before the receiver is ready to receive it.

B.2.7 Application Layer (3WI KFD Interface Protocol)

The KFD Interface Protocol is an application layer protocol that sits on top of the 4kbps Transfer Protocol for Three Wire Half Duplex interface. Its purpose is to provide a method for exchanging Key Management Messages (or KMMs) on the KFD interface.

The KFD Interface Protocol defines the format of data and control frames on the KFD-MR interface.

Data frames are used to deliver KMMs and are referred to as KMM frames. The format of a KMM frame supports the following capabilities:

- Allows for variable sized KMMs to be transported across the KFD-MR interface.
- Allows for encryption of KMMs on the KFD-MR interface. The encryption format for KMMs is defined in reference [1].
- Allows for detection of transmission errors on the KFD-MR interface.

Control frames are generated by the protocol and contain no data. These frames are primarily used to facilitate the exchange of KMM frames. All control frames are a single byte in length.

The KFD Interface Protocol also defines the procedure for the exchange of KMMs between the KFD and MR. This exchange procedure is referred to as an exchange session. The exchange session takes into account the half-duplex nature of the physical interface when using Three Wire Half Duplex by requiring that the direction of data flow alternate. After a device sends a frame on the interface, it shall go into receive to accept a frame. Only then can it send a subsequent frame. Furthermore, the exchange session utilizes control frames that identify the type of target that is connected to the KFD. This information can be used by the KFD to determine what capabilities the connected target has.

When using Three Wire Half Duplex, the KFD Interface Protocol takes KMM frames and control frames and hands off one byte at a time to the 4kbps Transfer Protocol for transfer across the KFD-MR interface. When the last byte has been handed off, the 4kbps Transfer Protocol is notified so the line can be turned around. Likewise, bytes received by the 4kbps Transfer Protocol are handed up to the KFD Interface Protocol. The KFD Interface Protocol parses the received bytes and identifies the received frames as KMM frames or control frames.

B.2.7.1 KFD Interface Protocol OPCODES

The KFD Interface Protocol frame consists of, at a minimum, an opcode field. Control frames contain only the opcode field while data frames contain additional data.

The complete list of opcodes, relevant to an OTAR implementation is as follows:

OPCODES	FRAME TYPE	USAGE
Ready Req [\$C0]	Control	Used by the KFD to request if a connected MR speaks KFD Interface Protocol.
Ready General Mode [\$D0]	Control	Sent in response to a Ready Req and indicates the MR speaks KFD Interface Protocol and is operating in a general environment or mode.
Transfer Done [\$C1]	Control	Indicates that a KFD or MR has transferred all queued KMMs.
KMM [\$C2]	Data	Indicates the subsequent octets are a KMM frame containing a KMM.
Disconnect ACK [\$90]	Control	Sent in response to a Disconnect and indicates that the MR has terminated the exchange session.
Disconnect [\$92]	Control	Indicates the transfer is complete and is being terminated

Table 25. KFD Opcode Field Values

B.2.7.2 KMM Frame

The KFD opcode is used to indicate that a KMM Frame follows immediately. This is the only data frame currently defined.

The KFD Interface Protocol wraps each KMM with a header as described in reference [1]. The header along with the KMM itself is referred to as a KMM Frame.

When transmitting, the first field in each message body is transmitted first, the most significant octet in each field is transmitted first, and the most significant bit in each octet is transmitted first.

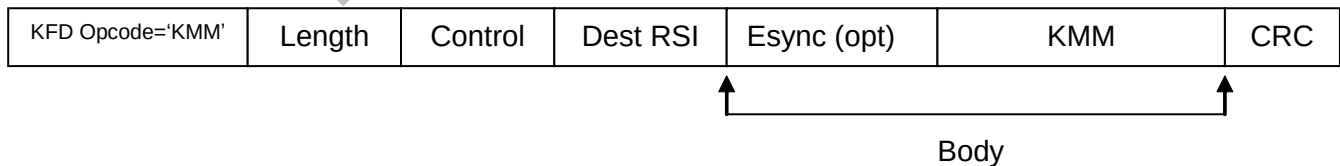


Figure 28. KMM Frame Header Format

The frame header format fields are described in the following sections.

B.2.7.2.1 Length

A two-octet field that identifies the length of the KMM Frame. The Control, Dest RSI, Body, and CRC fields are included in the length calculation.

Note: KMMs larger than 512 bytes shall not be sent.

B.2.7.2.2 Control

A one-octet field that contains a collection of control bits. The most significant bit of the Control field starts with b_7 and ends with b_0 as the least significant bit. If transmitted in the Three Wire Half Duplex 4kbps Transfer Protocol described in Section B.2.1, b_x shall be transmitted in the Dx slot for $x = 0..7$. See Figure 26.

Control Bit	Usage
b0:unavailable	This bit shall be set to 0.
b1:ENC: Encryption	%0: outer-layer encryption not applied %1: outer-layer encryption applied
b2:unavailable	This bit shall be set to 0.
b3:unavailable	This bit shall be set to 0.
b4:Reserved	This bit may be defined in the future and shall be set to 0.
b5:Reserved	This bit may be defined in the future and shall be set to 0.
b6:Reserved	This bit may be defined in the future and shall be set to 0.
b7:Reserved	This bit may be defined in the future and shall be set to 0.

Table 26. KMM Frame Control Bit Definitions

B.2.7.2.3 Destination RSI

A three octet field containing the Radio Set Identifier of the secure unit to which the KMM is addressed.

B.2.7.2.4 Body

A variable length field containing a KMM. If outer layer encryption is used, the body contains an esync block as well as the KMM. The encryption format for KMMs is defined in reference [1].

B.2.7.2.4.1 Body-Esync Block Format

The esync block contains the information needed to outer layer decrypt the KMM contained in the KMM Frame. It consists of the following fields:

- 9 byte Message Indicator
- 1 byte algorithm ID
- 2 byte key ID

1 byte secondary SAP (used as a place holder for consistency with CAI format; always set to \$00)

B.2.7.2.4.2 KMM

This field contains one of the KMMs defined in section 3.9 of this document.

B.2.7.2.5 CRC

A two-byte field containing CRC calculated over the Control, Dest RSI, and Body fields. The CRC is calculated using the CRC-CCITT 16 bit polynomial $x^{16} + x^{12} + x^5 + 1$. The state of the input register is initialized to all ones. See Annex A for more details.

B.2.7.3 Exchange Procedure

This section defines the exchange procedure, referred to as an exchange session, that is used by the KFD Interface protocol. The protocol divides the exchange session into four parts. Refer to Figure 29.

- 1) Protocol selection handshake
- 2) KMM transfer from KFD to MR
- 3) KMM transfer from MR to KFD
- 4) Disconnect (termination of transfer)

Step 1: The protocol selection handshake is used to inform the MR of which protocol is used for the exchange session, in this case the KFD Interface. The KFD sends a *Ready Req.* If the MR is capable of exchanging information using the requested protocol, it responds with a *Ready General Mode*, else it does not respond. If no response or an invalid response is received, the KFD times out and terminate the connection. Subsequent communications start with a new key signature. If the proper response is received, then the exchange session proceeds to step 2.

Step 2: The KFD now transfers any and all KMM frames queued for this MR. To minimize the memory requirements of the MR, the KFD transfers the KMM frames one at a time waiting for a reply after each transfer. The MR replies with a KMM frame containing the appropriate response KMM. Once all KMMs have been transferred, the KFD sends a Transfer Done indicating to the MR that it has completed its transfer. The exchange session now moves to step 3.

Step 3: The MR now transfers any and all KMMs queued for the KFD in the same manner as step 2. Once all KMMs have been transferred, the MR sends a Transfer Done to indicate to the KFD that it has completed.

Note: Typically, step 3 should not be required since the MR does not have any KMMs for the KFD. It simply sends a Transfer Done.

Step 4: The KFD now terminates the connection by sending a Disconnect to the MR. The MR sends a Disconnect Ack in response to the Disconnect opcode from the KFD.

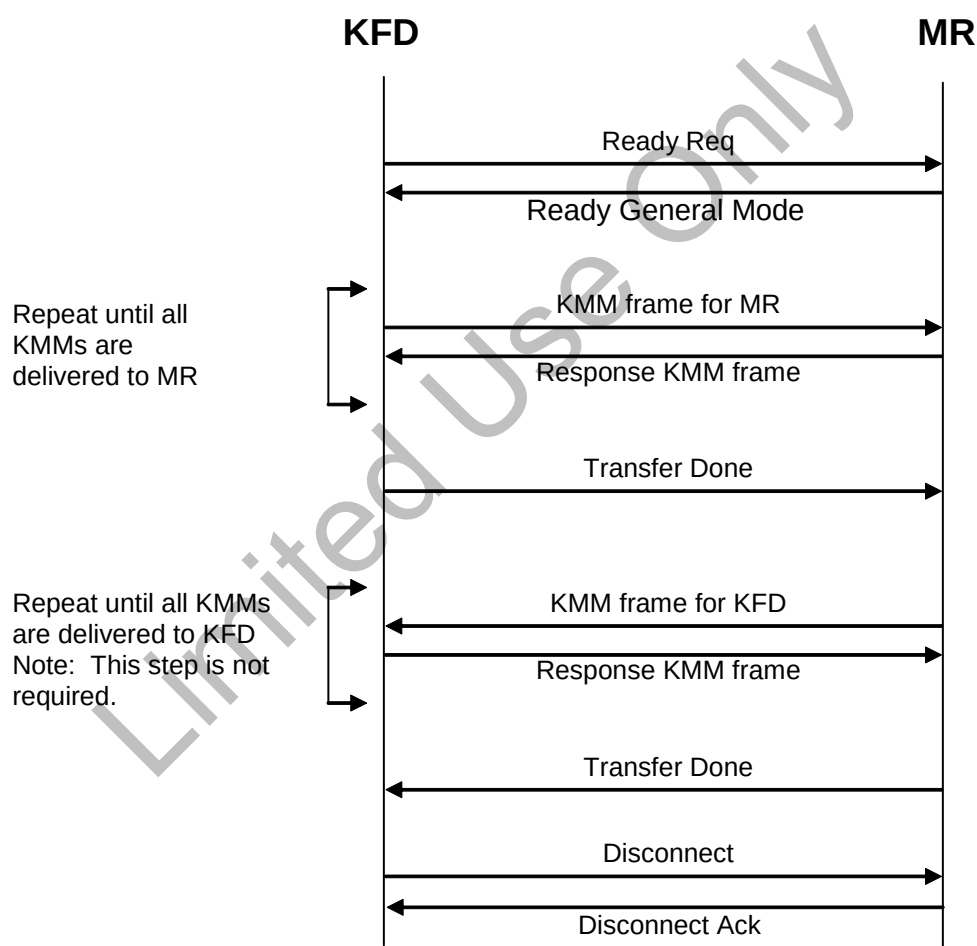


Figure 29. 3WI Exchange Session defined by the KFD- Interface Protocol

While physically connected to the MR, a KFD can initiate multiple exchange sessions. Each exchange session is preceded by a Ready Req and consists of

the four steps described above. The procedures in all four steps shall follow the timeout considerations that are defined in section B.2.4.

A KFD may also execute an exchange session without actually transferring any KMMs. This would be done for the purpose of determining if the connected MR is capable of exchanging information using the KFD Interface Protocol. Such an exchange session would consist of the following frames being exchanged:

1. Ready Req from KFD to MR
2. Ready General Mode from MR to KFD
3. Transfer Done from KFD to MR
4. Transfer Done from MR to KFD
5. Disconnect from KFD to MR
6. Disconnect Ack from MR to KFD

From this exchange session, the KFD could determine that a MR was connected.

B.2.8 Application Layer (Manual Rekeying Application)

The Manual Rekeying application provides the manual rekeying KFD features. The Manual Rekeying application determines what KMMs need to be transferred for the particular KFD feature selected. The KMMs are then passed to KFD Interface Protocol for transfer to the MR.

The following sections identify the manual rekeying procedures that are used to implement the manual rekeying features. A manual rekeying procedure consists of a KMM being sent from the KFD to the MR and the appropriate KMM response being sent from the MR to the KFD. For some manual rekeying features, a series of manual rekeying procedures is required.

Furthermore, the MR shall respond with a Negative Acknowledgement KMM (Status = \$03 Invalid Message ID) for all messages that the MR does not understand.

B.2.8.1 Keyload

To load encryption keys, the following KMM exchange is recommended:

1. The KFD and MR establish a connection.
2. The KFD sends an Inventory-Command (List Active Keyset IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD sends one or more Modify-Key-Command KMMs to the MR. The MR replies to each with a Rekey-Ack KMM.
4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Keyload feature is shown below.

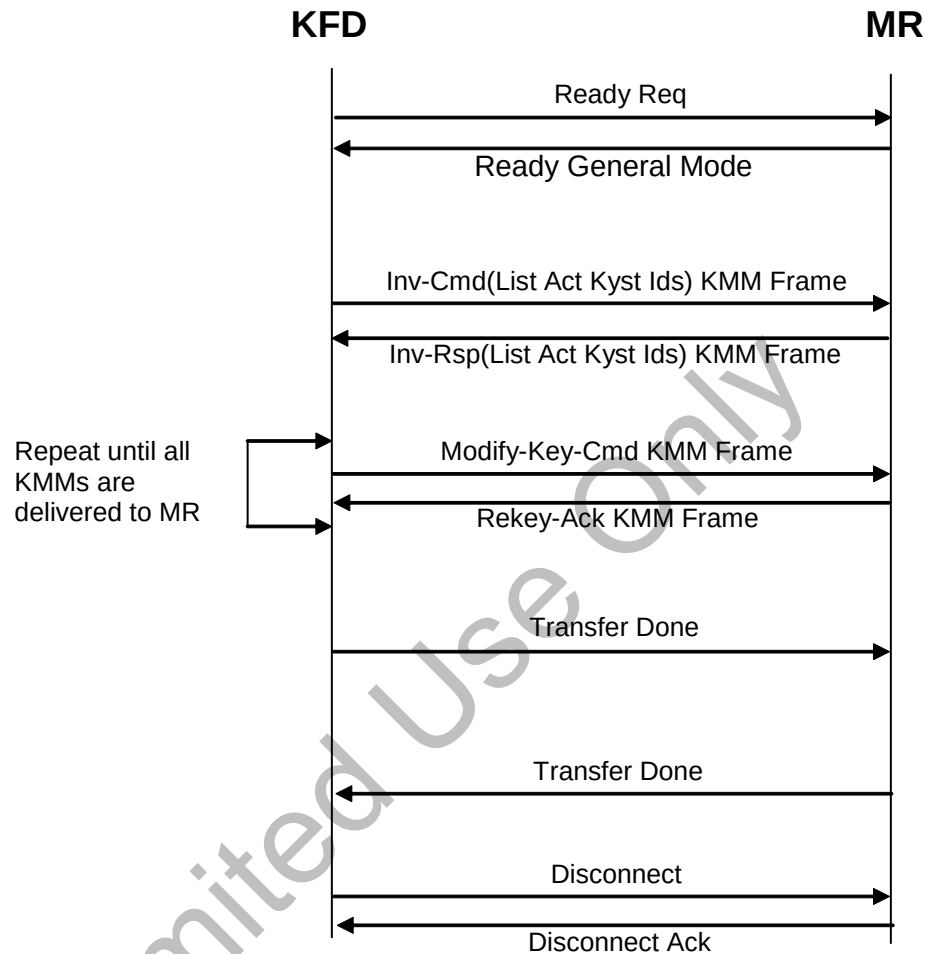


Figure 30. 3WI Exchange Session for the Keyload feature

The keyload procedure supports the loading of multiple keys. Multiple keys of the same algorithm can be contained in a single Modify Key Command KMM. The size of the KMM is restricted to a predetermined length (see section 3.9). The KFD repeats the Modify Key Command exchange until all keys are transferred.

Special mention is given to how keysets are initialized in the MR. The necessary operations are described as follows:

1. The MR has no keysets enabled or activated when first brought into service. An Inventory query issued by the KFD results in an indication of no keysets being found.

2. The first keyset of any order within a crypto group that is populated with keys from the KFD are automatically enabled and activated.
3. If a new keyset in a crypto group is populated with keys, and there already is an active and enabled keyset in the crypto group, the new keyset is enabled but not activated. The currently active keyset remains active until a Changeover procedure changes its activation state. There is no scenario where the only enabled keyset in a crypto group is not active.

New keysets are enabled by use of the Modify Key command.

B.2.8.2 Load Individual RSI

To load an individual RSI into a connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List RSI Items) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD sends a Change RSI Command KMM to the MR. The MR replies with a Change RSI Response KMM.
4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Load the Individual RSI feature is shown below.

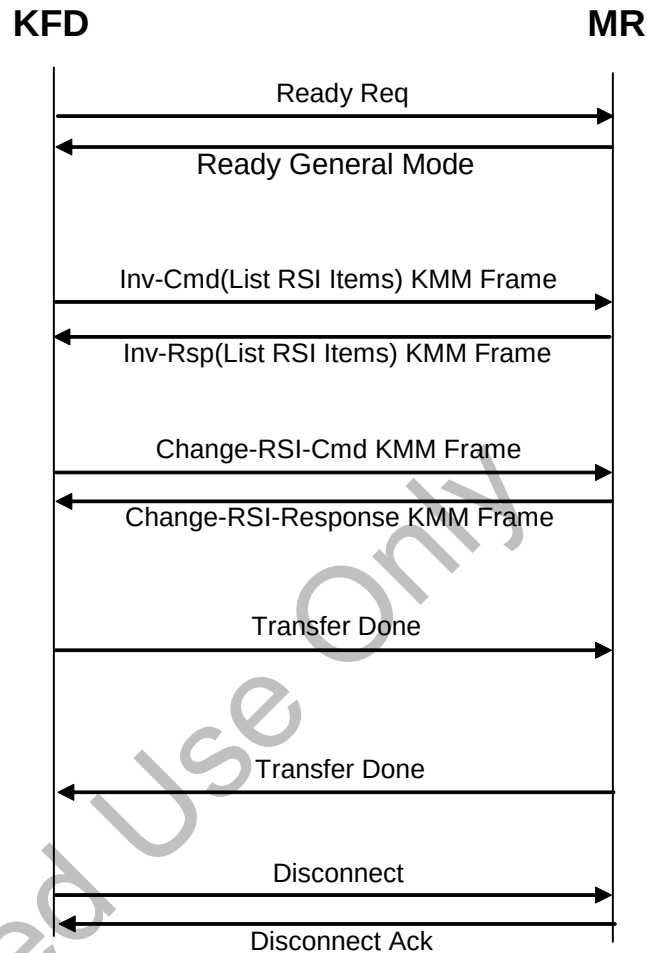


Figure 31. 3WI Exchange Session for the Load Individual RSI feature

The Message Number field within the KFD version of Change RSI Command KMM shall always be set to zero. No other values are allowed, nor should they be accepted by the MR. The MR shall correspondingly set its inbound and outbound Message Numbers to zero upon receiving the Change RSI Command KMM. This requirement applies to any RSI, Individual or Group, to which a Change RSI Command originating from the KFD was targeted.

B.2.8.3 Load KMF RSI

To program the proper the KMF RSI into a connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List MNP) KMM to the MR. The MR replies with an Inventory-Response KMM.

- 3. The KFD sends a Load-Config Command KMM with the MNP from the MR and the new KMF RSI. The MR replies with a Load-Config-Response KMM.
- 4. The KFD and MR complete the transaction.
- 5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Load KMF RSI feature is shown below.

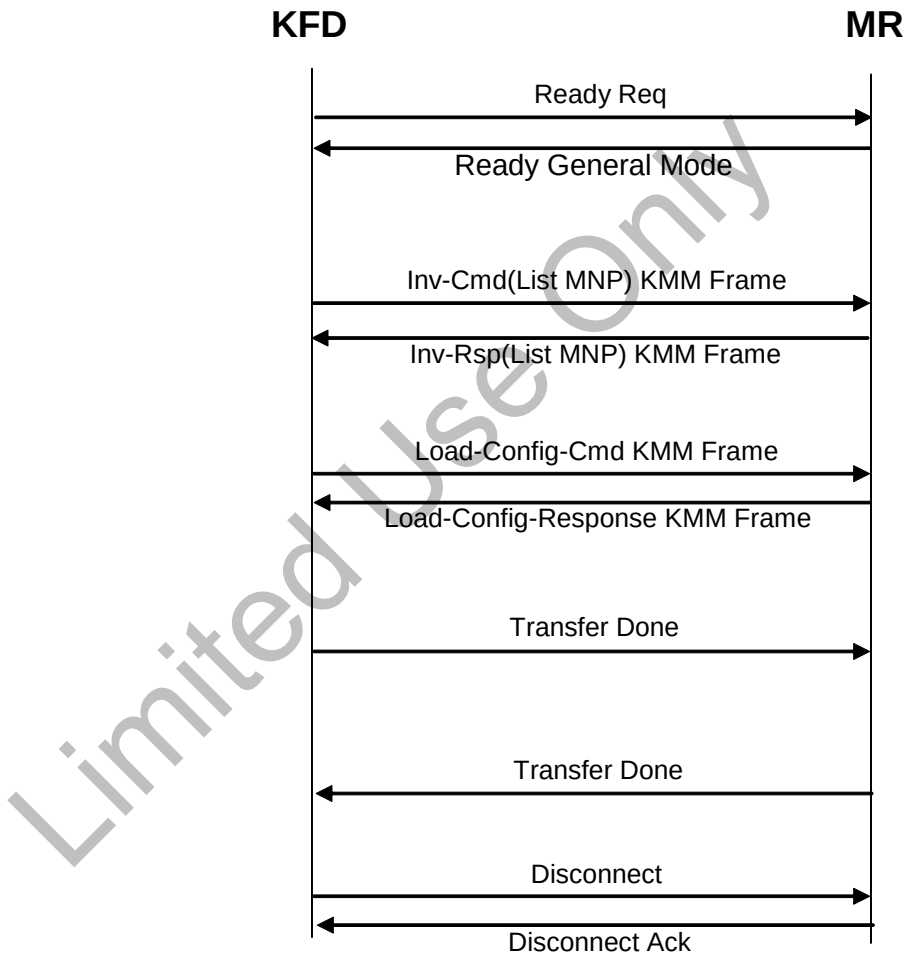


Figure 32. 3WI Exchange Session for the Load KMF RSI feature

B.2.8.4 Load MNP

To program the proper Message Number Period into a connected MR, the following KMM exchange is recommended:

- 1. The Key Fill Device and MR establish a connection.

2. The KFD sends an Inventory-Command (List KMF RSI) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD sends a Load-Config Command KMM with the KMF RSI from the MR and the new MNP. The MR replies with a Load-Config-Response KMM.
4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Load MNP feature is shown below.

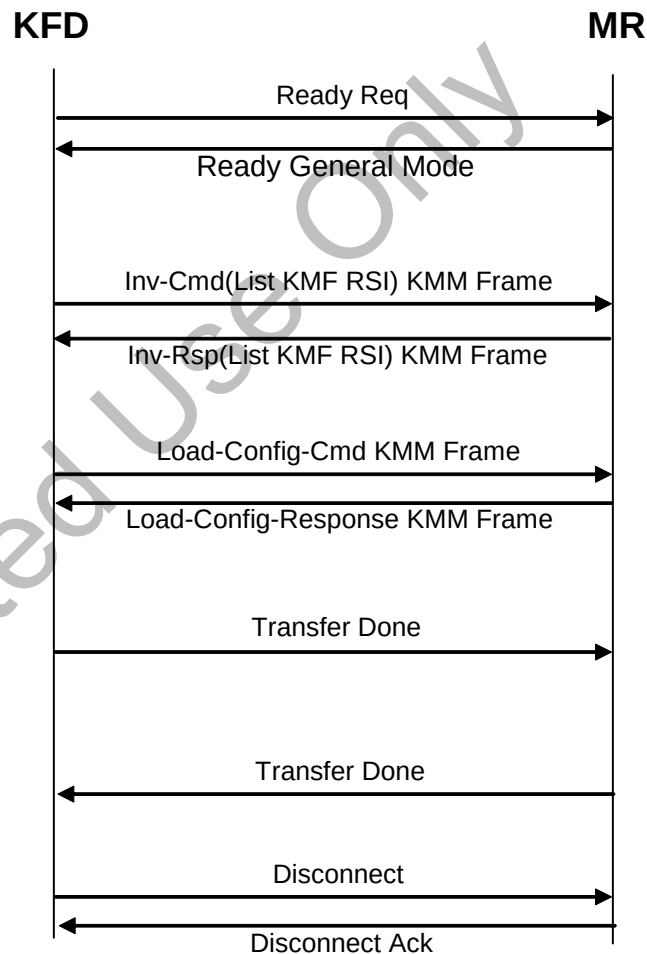


Figure 33. 3WI Exchange Session for the Load MNP feature

B.2.8.5 Erase Keys

To erase keys in a connected MR, the following KMM exchange is recommended:

1. The KFD and MR establish a connection.
2. The KFD sends an Inventory-Command (List Active Keyset IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD sends a Modify-Key-Command KMM identifying the key to be erased. Only the key in the specified SLN and keyset is erased. The Delete/Rekey bar of the Key Format primitive field is set to %1 in this case. The MR replies to each with a Rekey-Ack KMM.
4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Erase Keys feature is shown below.

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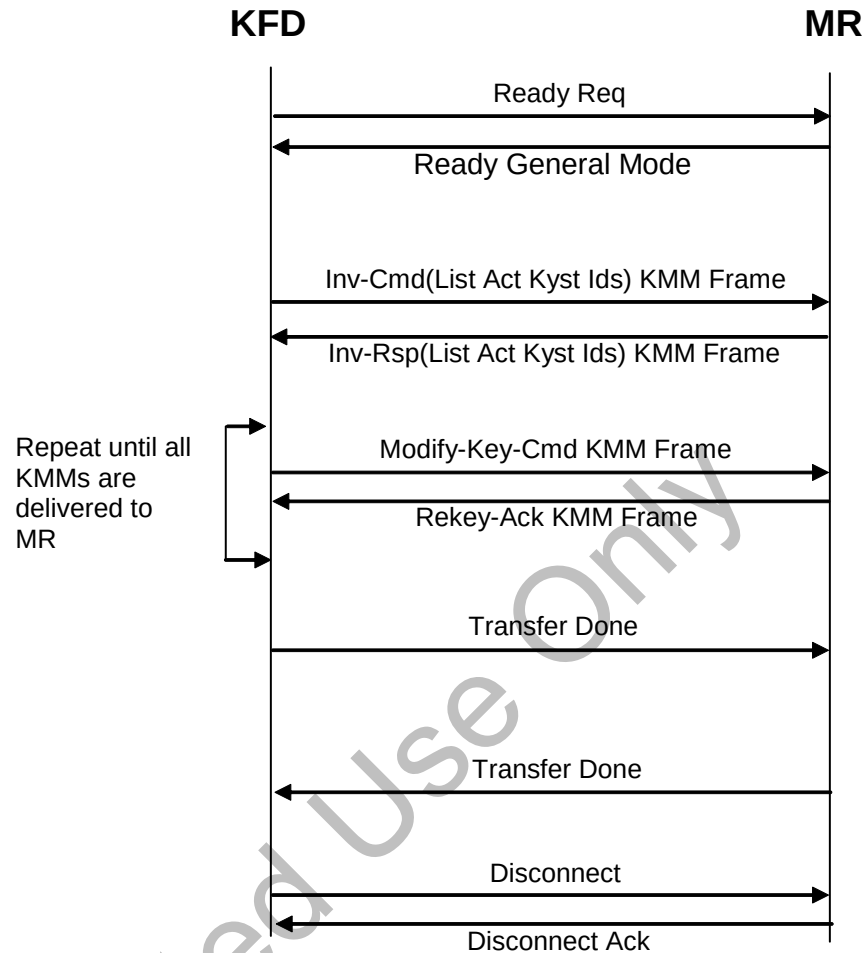


Figure 34. 3WI Exchange Session for the Erase Keys feature

The key erase procedure supports the deletion of multiple keys. The deletion commands for multiple keys can be contained in a single Modify Key Command KMM. The size of the KMM is restricted to a predetermined length (see 3.9). The KFD repeats the Modify Key Command exchange until all key deletion notifications are transferred.

B.2.8.6 Erase All Keys

To erase all keys in a connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends a Zeroize-Command KMM to the MR. The MR replies with a Zeroize-Response KMM.
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Erase All Keys feature is shown below.

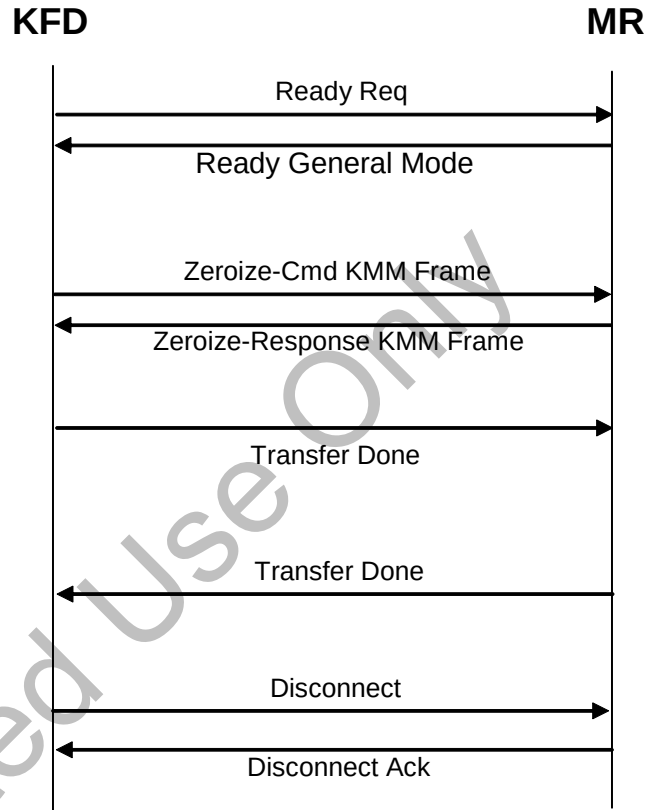


Figure 35. 3WI Exchange Session for the Erase All Keys feature

B.2.8.7 View Key Info

To view information about the keys contained in a connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List Active Keys) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. Repeat step 2 if the number of keys exceeds what can be returned in a single KMM.

4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View Key Info feature is shown below.

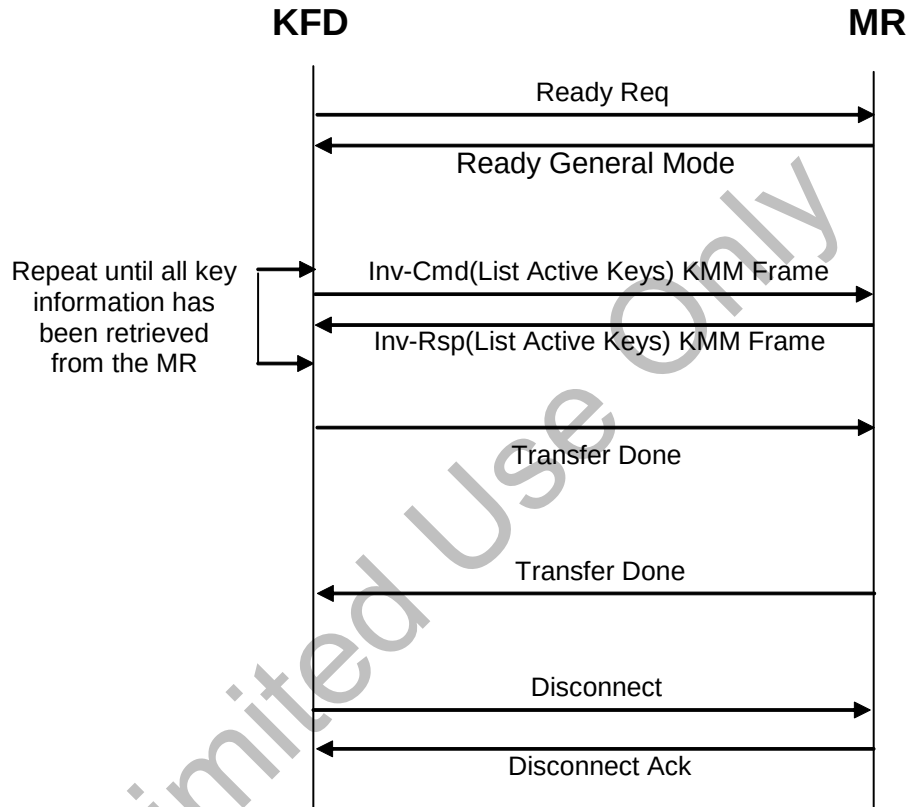


Figure 36. 3WI Exchange Session for the View Key Info feature

B.2.8.8 View Individual RSI

To view the individual RSI of the connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List RSI Items) KMM to the MR. The MR replies with an Inventory-Response KMM.

- 3. The KFD and MR complete the transaction.
- 4. The KFD and the MR remove the connection.

Note: The first RSI returned should be the individual RSI.

The complete exchange session showing all KFD Interface Protocol frames for the View Individual RSI feature is shown below.

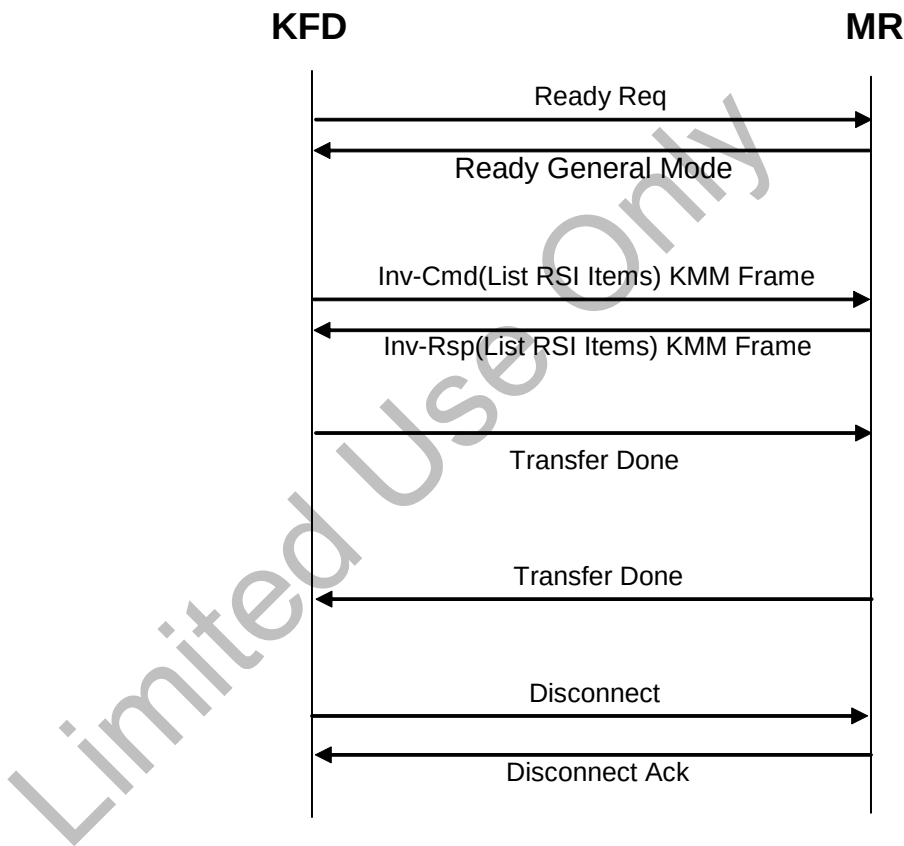


Figure 37. 3WI Exchange Session for the View Individual RSI feature

B.2.8.9 View KMF RSI

To view the KMF RSI programmed in a connected MR, the following KMM exchange is recommended:

- 1. The Key Fill Device and MR establish a connection.

2. The KFD sends an Inventory-Command (List KMF RSI) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View KMF RSI feature is shown below.

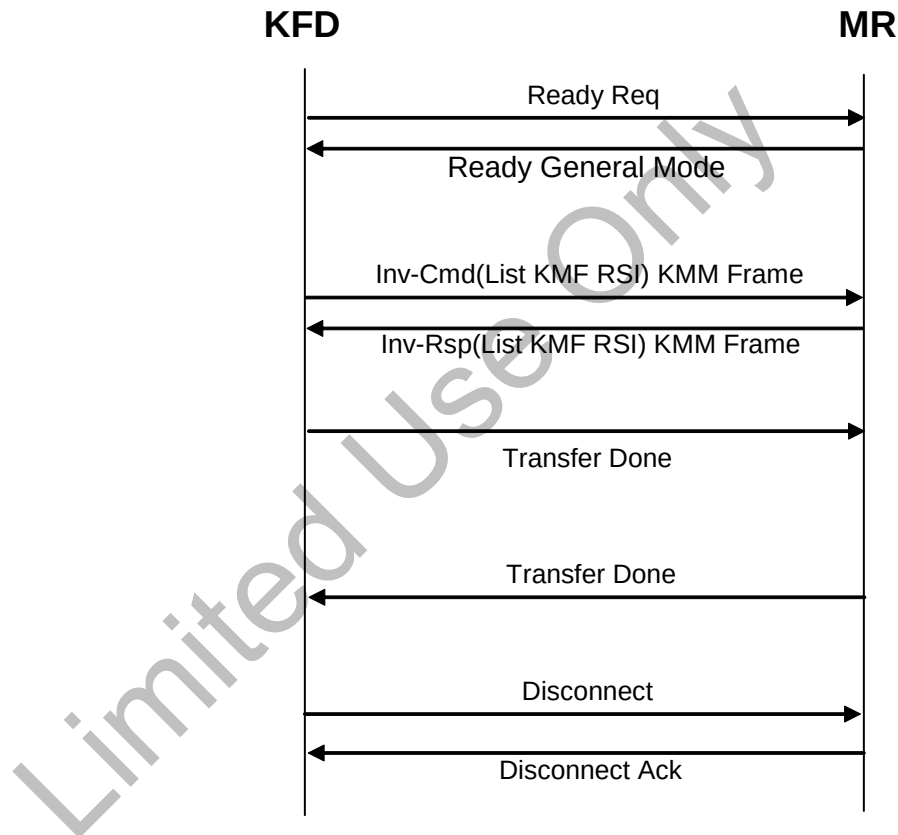


Figure 38. 3WI Exchange Session for the View KMF RSI feature

B.2.8.10 View MNP

To view the Message Number Period programmed in the connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.

- 2. The KFD sends an Inventory-Command (List MNP) KMM to the MR. The MR replies with an Inventory-Response KMM.
- 3. The KFD and MR complete the transaction.
- 4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View MNP feature is shown below.

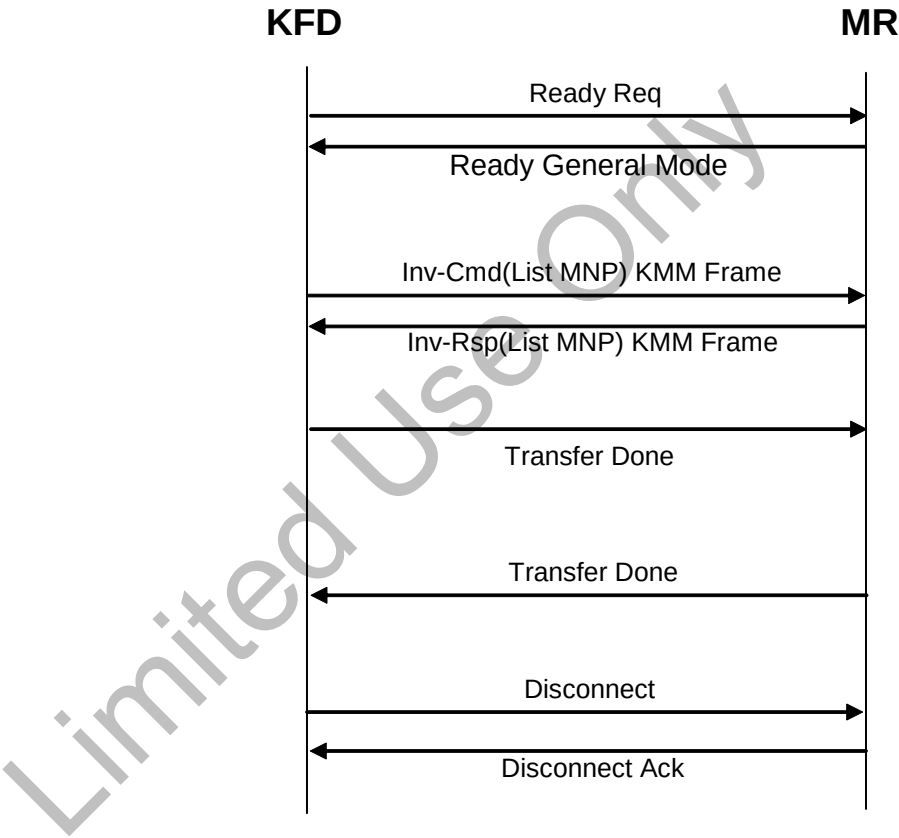


Figure 39. 3WI Exchange Session for the View MNP feature

B.2.8.11 View Keypad Info

To view keypad information stored in a connected MR, the following KMM exchange is recommended:

- 1. The Key Fill Device and MR establish a connection.

2. The KFD sends an Inventory-Command (List Active Keyst IDs) KMM to the MR. The MR replies with an Inventory-Response KMM.
3. The KFD sends an Inventory-Command (List Keyst Tagging Info) KMM to the MR. The MR replies with an Inventory-Response KMM.
4. Repeat step 3 if the keyset information exceeds what can be returned in a single KMM.
5. The KFD and MR complete the transaction.
6. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View Keyst Info feature is shown below.

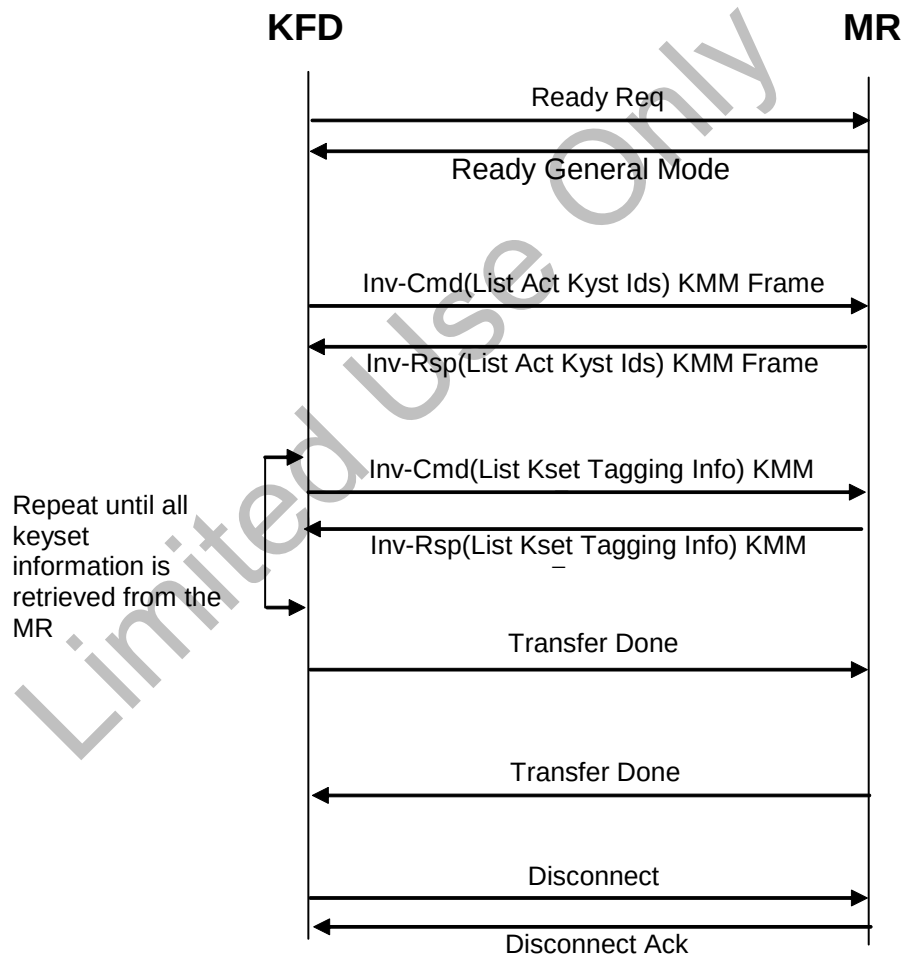


Figure 40. 3WI Exchange Session for the View Keyst Info feature

B.2.8.12 Activate Keypad

To activate a keypad in the connected MR, the following KMM exchange is recommended:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends a Changeover-Command KMM to the MR. The MR replies with a Changeover-Response KMM.
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Activate Keypad feature is shown below.

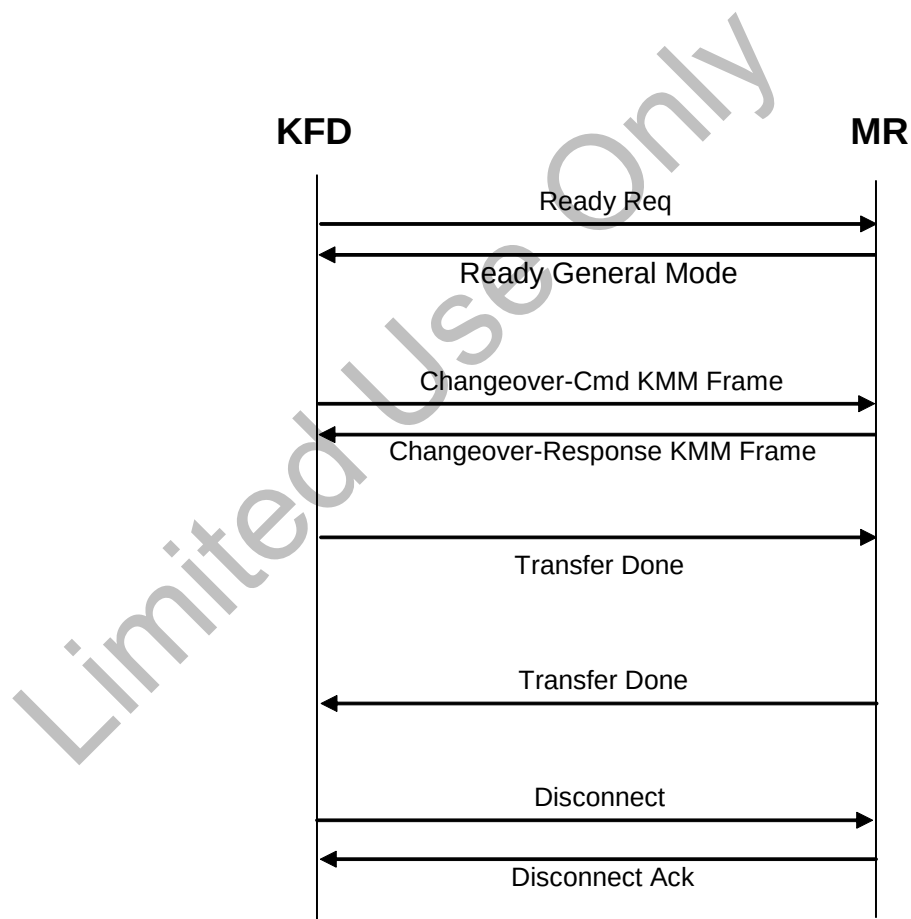


Figure 41. 3WI Exchange Session for the Activate Keypad feature

B.2.8.13 Load Authentication Key

To load an Authentication key into the connected MR, the following exchange is initiated:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends a Load Authentication Key -Command KMM to the MR, and the MR replies with a Load Authentication Key-Response KMM containing the SUID that was loaded with the authentication key.
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Load Authentication Key feature is shown below. Each run of the complete exchange session can load one authentication key. If multiple authentication keys need to be loaded, then the complete exchange session is run multiple times.

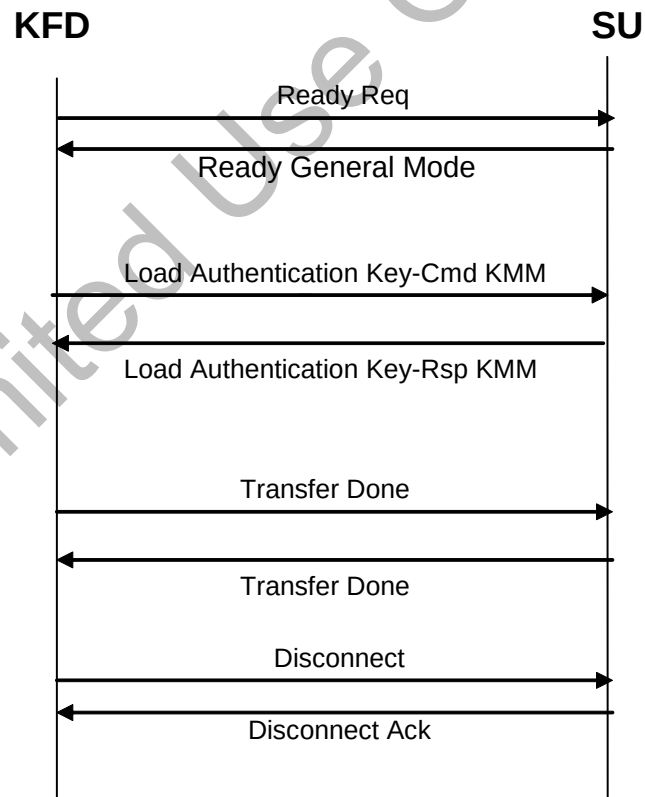


Figure 42. 3WI Exchange Session for the Load Authentication Key feature

B.2.8.14 Delete Authentication Key

To delete an authentication key from the connected MR, the following exchange is initiated:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends a Delete Authentication Key -Command KMM to the MR and the MR replies with a Delete-Authentication-Key-Response KMM.
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the Delete Authentication Key feature is shown below. If multiple authentication keys need to be deleted, then the complete exchange session is run multiple times.

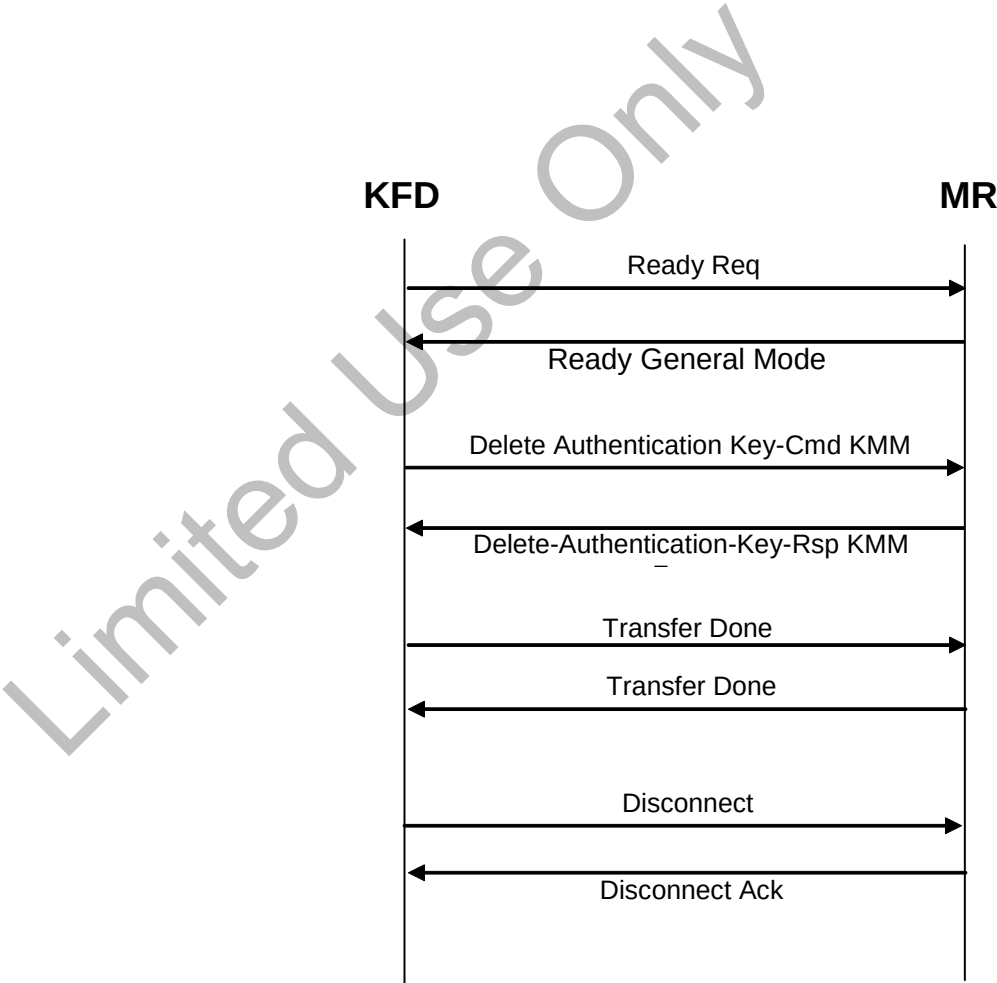


Figure 43. 3WI Exchange Session for the Delete Authentication Key feature

B.2.8.15 View SUID Info

To view information about the SUID contained in a connected MR, the following exchange is performed:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List SUID Items) KMM to the MR and the MR replies with an Inventory-Response KMM containing the SUIDs in the MR. (This step is repeated for a number of SUIDs that exceeds what can be returned in a single KMM).
3. The KFD and MR complete the transaction.
4. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View SUID Info feature is shown below.

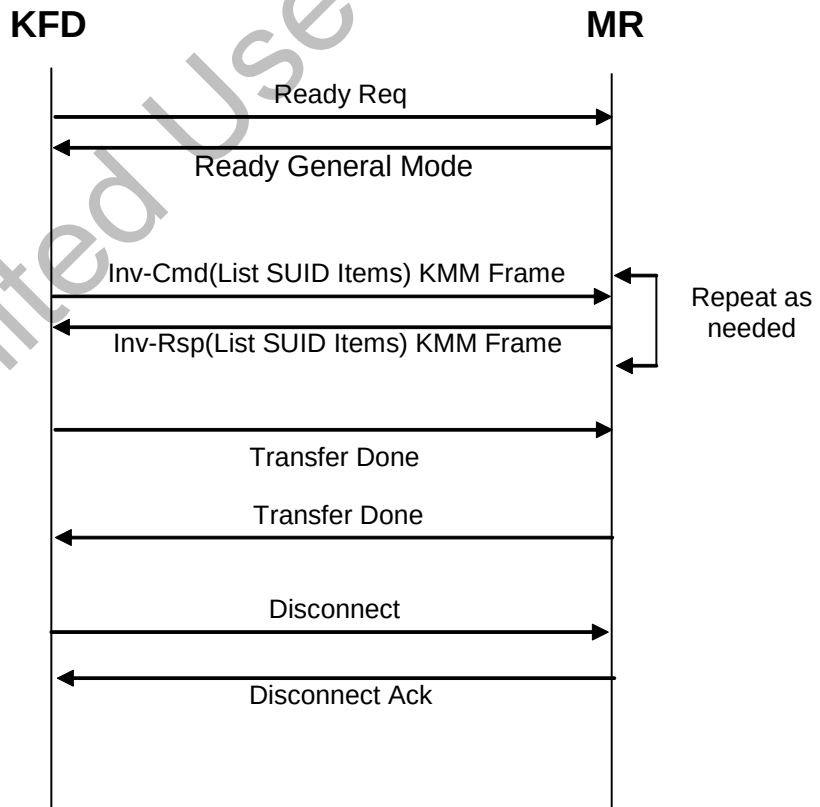


Figure 44. 3WI Exchange Session for the View SUID Info feature**B.2.8.16 View Active SUID Info**

To view information about the Active SUID in a connected MR and optionally load an Authentication key, the following exchange is performed:

1. The Key Fill Device and MR establish a connection.
2. The KFD sends an Inventory-Command (List Active SUID) KMM to the MR and the MR replies with an Inventory-Response KMM containing the Active SUID in the MR.
3. (Optional Step) The KFD sends a Load Authentication Key -Command KMM to the MR, and the MR replies with a Load Authentication Key-Response KMM containing the SUID that was loaded with the authentication key.
4. The KFD and MR complete the transaction.
5. The KFD and the MR remove the connection.

The complete exchange session showing all KFD Interface Protocol frames for the View SUID Info feature is shown below. Each run of the complete exchange session may load only one authentication key. If multiple authentication keys need to be loaded, then the complete exchange session is run multiple times.

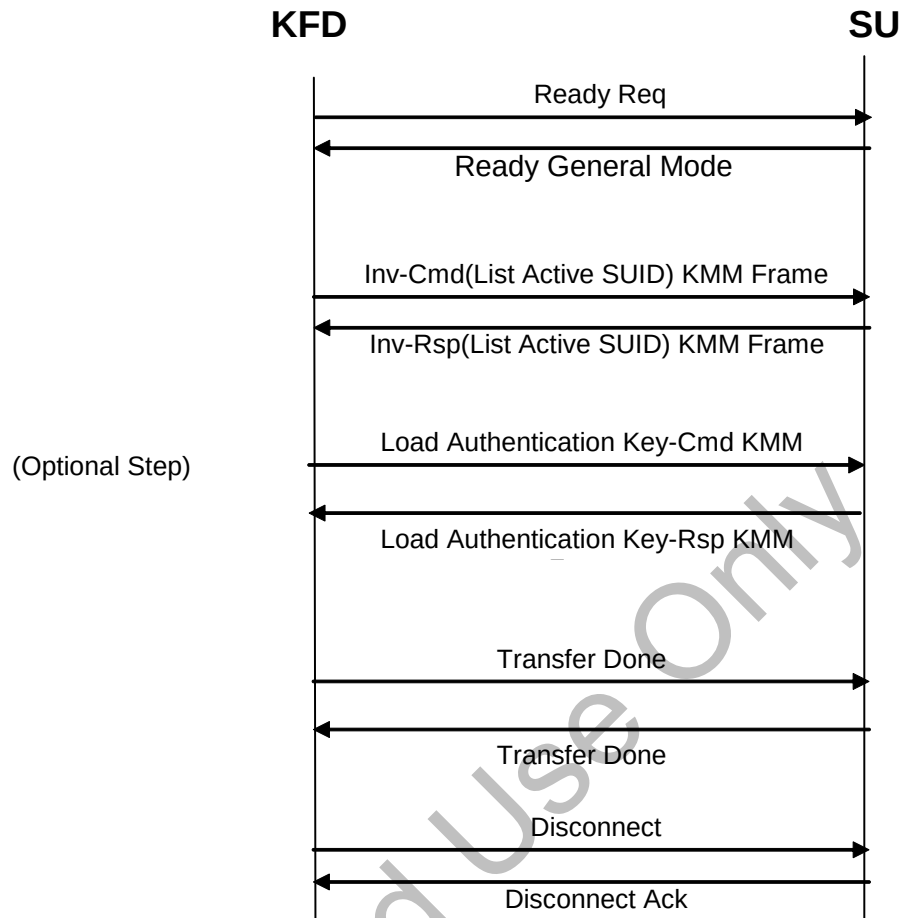


Figure 45. 3WI Exchange Session for the View Active SUID Info feature

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